

ROBOT ANALYSIS

The Mechanics of Serial and Parallel Manipulators

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PREFACE

This book has evolved from class notes used for a graduate-level course in robot analysis that the author has taught for over a decade at the University of Maryland, College Park, Maryland. It provides both fundamental and advanced topics on the kinematics, statics, and dynamics of robot manipulators. Both serial and parallel manipulators are covered in depth. In addition, it contains chapters on geared robotic mechanisms and tendon-driven manipulators. An overview of the state of the art in solving systems of nonlinear equations that arise in robot kinematics is given in the appendixes. An extensive reference list is provided at the end of each chapter to provide readers with plenty of resources for more advanced study.

Prerequisites for readers of this book consist of the fundamentals of kinematics, dynamics, vector space analysis, and matrix theory. These basics are usually taught at the undergraduate level. A one-semester course will cover most of the fundamentals presented in the book. The more advanced topics, such as the screw-based Jacobian and singularity analysis and the two chapters on geared robotic wrists and tendon-driven manipulators, can be taught in a second semester or be used as reference material. It is anticipated that after studying the kinematics and dynamics of robot manipulators, students will be motivated to take an advanced course in robot sensing, control, and machine intelligence.

Chapter 1 provides a brief review of the historical development and a classification of robot manipulators. The science of *robotics* and the *mechanics* of robot manipulators are defined. A brief review of some of the fundamentals that are essential for an understanding of the mechanics of robot manipulators

is given. Finally, the rotation matrix and homogeneous transformation matrix are described.

Chapter 2 lays the mathematical foundation for the position analysis of serial manipulators. The concept of loop-closure equation is introduced. Both the Denavit–Hartenberg method and the method of successive screw displacements are presented. Several industrial and research robots are analyzed to illustrate the methodologies.

Chapter 3 provides the fundamental knowledge needed for the position analysis of parallel manipulators. Parallel manipulators are classified according to their kinematic structures. The method of vector-loop equation is introduced. Several parallel manipulators, including a planar 3-dof manipulator, a spatial 3-dof orientation mechanism, a parallel manipulator with only translational degrees of freedom, and the well-known Stewart–Gough platform are analyzed.

Chapter 4 extends the study of serial manipulators from the position analysis to the velocity analysis. The differential kinematic properties of a link in an open-loop chain and their propagation from link to link are studied. Both the conventional Jacobian and the screw-based Jacobian are defined. The screw-based Jacobian analysis is shown to be more efficient than the conventional method of analysis. Finally, the singularities of serial manipulators are studied.

Chapter 5 covers the Jacobian and singularity analyses of parallel manipulators. The velocity vector-loop approach is employed for derivation of the conventional Jacobian matrices, and the concept of reciprocal screws is used for derivation of the screw-based Jacobian matrices. The singularities of parallel manipulators are classified into three types. For the first type of singularity, the manipulator loses 1 or more degrees of freedom, while for the second type of singularity, the manipulator gains 1 or more degrees of freedom. The third type of singularity occurs when the first and second types of singularity occur simultaneously. The physical meanings of each type of singularity are illustrated by several parallel manipulators.

Chapter 6 addresses the statics and stiffness of serial and parallel manipulators. Both the free-body diagram method and the method of virtual work are presented. By applying the principle of virtual work, it is shown that in the absence of gravity, the actuated joint torques are related to the end-effector output forces by the transpose of the Jacobian matrix. The statics and stiffness of several manipulators are analyzed to illustrate the principles.

Chapter 7 deals with the kinematics and statics of robotic wrist mechanisms. The structural characteristics of epicyclic gear trains are analyzed. The theory of fundamental circuits and the coaxiality conditions associated with epicyclic gear drives are derived. A systematic methodology for the kinematic analysis of epicyclic gear trains and robotic wrist mechanisms is developed.

Chapter 8 covers the kinematics and statics of tendon-driven manipulators. The concept of transmission lines is introduced. It is shown that tendon displacements are related to the joint angles by a structure matrix, and the structure matrix can be derived by inspection of the structural topology of the tendon routing. Furthermore, force transmission characteristics from the tendon space to the end-effector space are analyzed. An efficient method for the resolution of redundant tendon forces is developed. The Stanford/JPL hand and the Utah/MIT hand are analyzed to demonstrate the methodology.

Chapter 9 deals with the dynamics of serial manipulators. Both the recursive Newton–Euler formulation and the Lagrangian formulation are presented. The concepts of the Lagrangian function, manipulator inertia matrix, and generalized forces are introduced. Furthermore, the inertia effects of rotors on the dynamics of a serial manipulator are discussed.

Chapter 10 concerns the dynamics of parallel manipulators. The dynamical analysis of parallel manipulators is complicated by the existence of multiple closed loops. First, a numerical solution technique based on the laws of Newton and Euler is presented. Then a more efficient method based on the principle of virtual work is developed. Finally, it is shown that explicit equations of motion for some relatively simple manipulators can be derived by applying Lagrangian equations of the first type.

In the appendixes, the state-of-the-art continuation method and the methods of elimination for solving systems of nonlinear equations are presented. Furthermore, Raghavan and Roth's solution of the kinematics of the $6R$ manipulator of general geometry is presented.

Some of the materials presented in this book is based on the author's research in collaboration with friends, professional colleagues, and a former advisor: Ferdinand Freudenstein, Krishna Gupta, Tien Yien Li, Alexander Morgan, Madhusudan Raghavan, Bernard Roth, and Gregory Walsh—and with former students: Sun-Lai Chang, Dar-Zen Chen, Hsin-I Hsieh, Jyh-Jone Lee, Chen-Chou Lin, Yeong-Jeong Ou, Tai-Kang Shing, Richard Stamper, and Farhad Tahmasebi. Their efforts are deeply appreciated.

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CHAPTER 1

INTRODUCTION

Since the industrial revolution, there has been an ever-increasing demand to improve product quality and reduce manufacturing cost. At first, most products were manufactured by craftsmen. The product quality depended highly on the skill of the craftsmanship, and the product cost was prohibitively high. This is known as *manual production*.

At the beginning of the twentieth century (1905), Ford Motor Company introduced the concept of *mass production*. In a mass-production environment, most manufacturing processes are carried out by special-purpose machines. This method of production drastically reduces the cost of manufacturing and makes the automobile and other industrial products affordable to the general population. However, since each production machine is designed to perform a predetermined task, every time a new model is introduced, the entire production line has to be shut down and retooled. Retooling a production line during each model change can be very expensive. We call this type of automation *hard automation*.

The inflexibility and relatively high cost of hard automation have led to an entirely new approach. Recently, *robot manipulators* have been introduced by manufacturing industries for performing certain production tasks, such as material handling, spot welding, spray painting, and assembling. Because robot manipulators are controlled by computers or microprocessors, they can easily be reprogrammed for different tasks. Hence there is no need to replace these machines during a model change. We call this type of automation *flexible automation*.

There are compromises between manual production, hard automation, and flexible automation. In general, manual production tends to be slow and makes it difficult to achieve consistent product quality. Although hard automation is faster and more precise than the other two production methods, the retooling cost for each model change can be high. With flexible automation, on the other hand, much less time is required to set up a production line because robot manipulators can readily be purchased off the shelf. In addition, the manipulators can be preprogrammed off the line and are almost immune to obsolescence. Typically, when the production volume is very low, manual production is more economical. As the production volume increases, flexible automation becomes more cost-effective. As the production volume increases further, hard automation becomes the most effective production method. In this book, we study the mechanics of robot manipulators essential for the design, programming, and control of such devices in a flexible manufacturing environment.

1.1 HISTORICAL DEVELOPMENT

The word *robot* means different things to different people. *Webster's New World College Dictionary* defines a *robot* as "any anthropomorphic mechanical being built to do routine manual work for human beings." On the other hand, the definition used by the Robotics Institute of America is more specific: "A robot is a re-programmable multi-functional manipulator designed to move material, parts, tools, or specialized devices, through variable programmed motions for the performance of a variety of tasks." The latter definition includes mechanical manipulators, numerical control machines, walking machines, and the humanoids of science fiction. Although most people perceive robots anthropomorphically, today's industrial robots are only somewhat humanoid in appearance. In fact, most industrial robots are mechanical manipulators. Figure 1.1 shows a typical humanoid of science fiction; by contrast, Fig. 1.2 shows a typical industrial robot.

Although the word *robot* entered the English vocabulary as early as 1923 (Capek, 1923), the actual development of robot manipulators did not occur until after the 1940s. The emergence of industrial robots was due initially to the need to handle hazardous materials and in space exploration, and later to achieve flexible automation. During the late 1940s, the Oak Ridge and Argonne National Laboratories began their development of *master-slave manipulators* (Fig. 1.3) for handling radioactive materials. The master manipulator is guided by an operator, while the slave manipulator duplicates the motion produced by the master in a remote site. Force feedback can be incorporated



FIGURE 1.1. Humanoid. (Courtesy of SARCOS, Salt Lake City, Utah.)

to provide the operator with some feedback on the load the slave manipulator encountered.

In the late 1950s, George Devol conceived a programmable mechanical manipulator, subsequently refined by him together with Engelberger into the Unimate industrial robot (Engelberger, 1980). An important feature of an industrial robot is the incorporation of a computer- or microprocessor-based

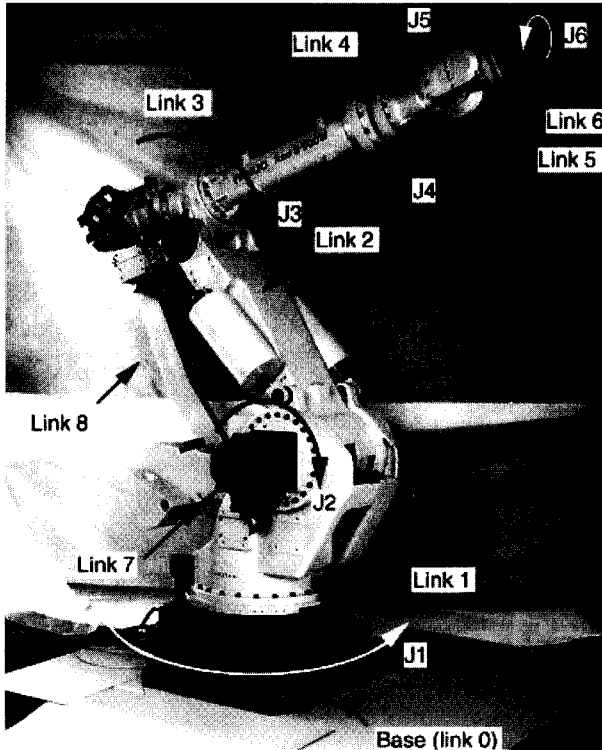


FIGURE 1.2. Fanuc S-900W robot. (Courtesy of Fanuc Robotics North America, Inc., Rochester Hills, Michigan.)

controller with sensory feedback to achieve programmable multifunctional capability. In the 1960s, the Stanford Artificial Intelligence Laboratory (McCarthy et al., 1968) and MIT Lincoln Laboratory (Ernst, 1962) began their development of robot manipulators with tactile sensors and computer vision. The Stanford robot was capable of doing some simple assembly tasks. In 1974, Cincinnati Milacron introduced a heavy-duty industrial robot called the Cincinnati Milacron T³ (The Tomorrow Tool) robot. Later, Unimation Incorporated introduced a new series of robots called the *PUMA robots*. Meanwhile, other companies and countries also began robotics research and development activities.

In the 1980s, there was a big boom in the robotics industry, due to the huge demand for more economical and flexible automation from the automotive industry and for space exploration. In 1982, General Motors, the largest user of robots, signed an agreement with Fanuc Ltd. to form a robotic company called GMF in Rochester Hills, Michigan, to produce robots for use primarily in the automotive industry. During the same period, many other U.S.

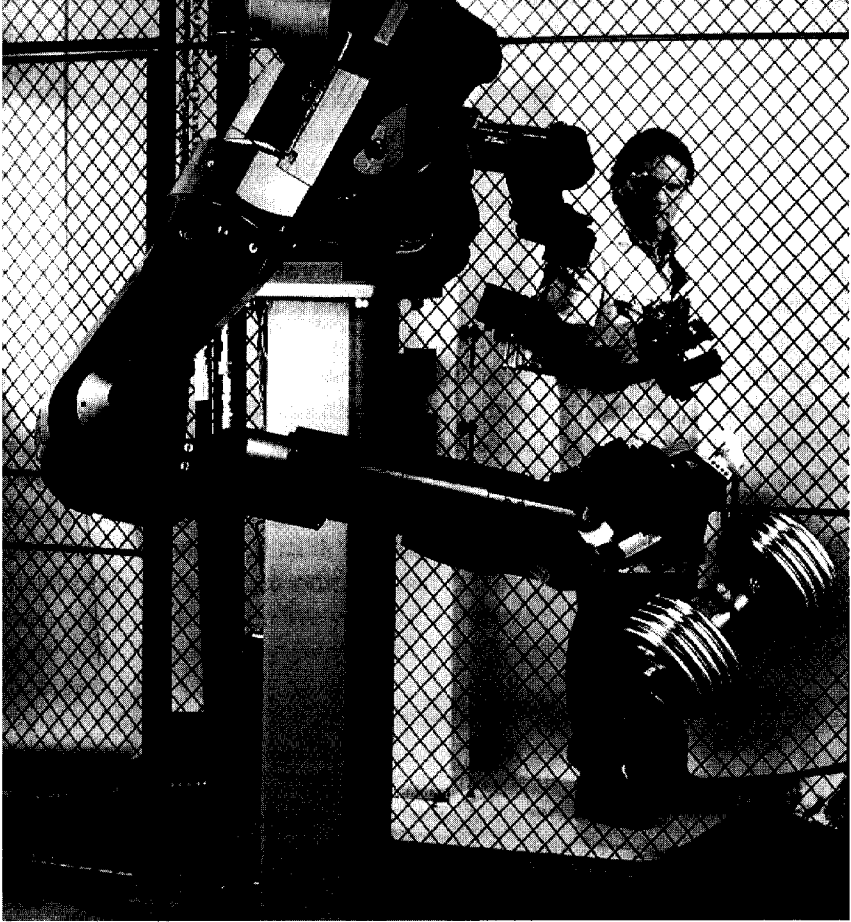


FIGURE 1.3. Master-slave robot. (Courtesy of SARCOS, Salt Lake City, Utah.)

companies, including Adept Technology, Bendix, General Electric, IBM, Intelledex, SARCOS, Westinghouse, and Zebra Robotics, also ventured into the business, but only a few were successful. In the early 1990s, GMF was sold back to Fanuc Ltd. and renamed Fanuc Robotics North America, Inc. Currently, there are more European and, in particular, Japanese robotic companies than U.S. firms. The leading Japanese robotic companies include such industrial giants as Fanuc, Hitachi, Kawasaki Robotics, Mitsubishi Electric, and Seiko Instruments. Some European companies, such as ASEA and Kuka International, also play an important role in both research and development and in marketing robot manipulators. Recently, Honda Laboratory invested

nearly \$80 million in the development of a 200-kg humanoid that is capable of walking up and down stairs. In 1997, NASA sent a rover called *Pathfinder* to Mars. A historical review of the early development of robot manipulators is given in Engelberger (1980).

During recent decades, robot manipulators have been used primarily for repetitive operations and in hazardous environments. Typical repetitive operation tasks include materials handling, parts loading and unloading, sealant application, and component assembling. Applications in hazardous environments include radioactive material handling, space and undersea exploration, spot welding, spray painting, and so on. Other robotic applications include construction robots, flight and vehicle simulators (Stewart, 1965), mining machines (Arai et al., 1991), pointing devices (Gosselin and Hamel, 1994), and intelligent machining centers (Giddings & Lewis, 1995). Recently, there has been increasing interest in making robots more intelligent and user friendly, and medical surgery robots, household service robots, and so on, are becoming available. Although much effort has been spent on the development of robot manipulators, the ultimate goal of developing intelligent, user-friendly robots that can emulate human functions is still in its infancy. Major obstacles can be attributed to the fact that some of the key technologies have not yet been fully developed.

1.2 A SENSE OF MECHANISMS

A mechanism or mechanical manipulator is made up of several links connected by joints. The number of degrees of freedom (dof) of a mechanism depends on the number of links and joints and the types of joints used for construction of the mechanism. In this section we define links, joints, kinematic chains, mechanisms, and machines, then introduce the concepts of degrees of freedom and the loop mobility criterion.

1.2.1 Links and Joints

The individual bodies that make up a mechanism are called the *members* or *links*. In this book, unless stated otherwise, we treat all links as rigid bodies. The assumption of rigid bodies makes the study of mechanisms and robot manipulators a lot easier to understand. However, for high-speed or highly loaded mechanisms, the elastic effects of a material body may become significant and should be taken into consideration. For convenience, certain non-rigid bodies, such as chains, cables, or belts, which momentarily serve the same function as rigid bodies, may also be considered as links. From the kinematic point of view, two or more members connected together such that

no relative motion can occur between them will be considered as a single link. For example, an assembly of two gears connected by a common shaft is treated kinematically as one link.

Links in a mechanism or mechanical manipulator are connected in pairs. The connection between two links is called a *joint*. A joint provides some physical constraints on the relative motion between the two members. The kind of relative motion permitted by a joint is governed by the form of the contact surfaces between the members. The contact surface of a link is called a *pair element*. Two pair elements form a *kinematic pair*. Kinematic pairs can be classified into lower pairs and higher pairs, according to the type of contact. A kinematic pair is called a *lower pair* if the two elements contact each other with a substantial surface area. Typically, the forms of two mating lower-pair elements are geometrically identical, one being solid and the other hollow. A kinematic pair is called a *higher pair* if the pair elements are in contact at a point or along a line. There are six basic types of lower pairs and two of higher pairs that are frequently used in mechanisms and robot manipulators.

A *revolute joint*, R , permits two paired elements to rotate with respect to each other about an axis that is defined by the geometry of the joint. Hence the revolute joint imposes five constraints between the paired elements and is a 1-dof joint. The revolute joint is sometimes called a *turning pair*, a *hinge*, or a *pin joint*.

A *prismatic joint*, P , allows two paired elements to slide with respect to each other along an axis that is defined by the geometry of the joint. Hence the prismatic joint imposes five constraints between the paired elements and is a 1-dof joint. The prismatic joint is sometimes called a *sliding pair*.

A *cylindrical joint*, C , permits rotation about, and independent translation along, an axis that is defined by the geometry of the joint. Hence the cylindrical joint imposes four constraints on the paired elements and is a 2-dof joint.

A *helical joint*, H , allows two paired elements to rotate about, and translate along, an axis defined by the geometry of the joint. However, the translation is related to rotation by the pitch of a screw. Hence the helical joint imposes five constraints on the paired elements and is a 1-dof joint. The helical joint is sometimes called a *screw pair*.

A *spherical joint*, S , allows one element to rotate freely with respect to the other about the center of the sphere in all possible orientations. No translation between the paired elements is permitted. Hence the spherical joint imposes three constraints on the paired elements and is a 3-dof joint.

A *plane pair*, E , permits two translational degrees of freedom along the plane of contact and a rotational degree of freedom about an axis normal to

the plane of contact. Hence it imposes three constraints on the paired elements and is a 3-dof joint.

A *gear pair*, G , permits one gear to roll and slide with respect to the other at the point of contact between two meshing teeth. In addition, the motion space of a spur, helical, or bevel gear is also assumed to be constrained on a plane. Hence a gear pair imposes four constraints on the paired elements and is a 2-dof joint.

A *cam pair*, C_p , is similar to the gear pair except that a spring is usually used to keep the two paired elements in contact. Hence the cam pair is a 2-dof joint.

Revolute, prismatic, cylindrical, helical, spherical, and plane pairs are lower pairs. Gear and cam pairs are higher pairs. Another frequently used joint is the *universal joint*, shown in Fig. 1.4. The universal joint is essentially a combination of two intersecting revolute joints, hence is a 2-dof joint.

1.2.2 Kinematic Chains, Mechanisms, and Machines

A *kinematic chain* is an assemblage of links that are connected by joints. When every link in a kinematic chain is connected to every other link by at least two distinct paths, the kinematic chain forms one or more closed loops and is called a *closed-loop chain*. On the other hand, if every link is connected to every other link by one and only one path, the kinematic chain is called an *open-loop chain*. It is also possible for a kinematic chain to be made up of both closed- and open-loop chains. We call such a chain a *hybrid kinematic chain*.

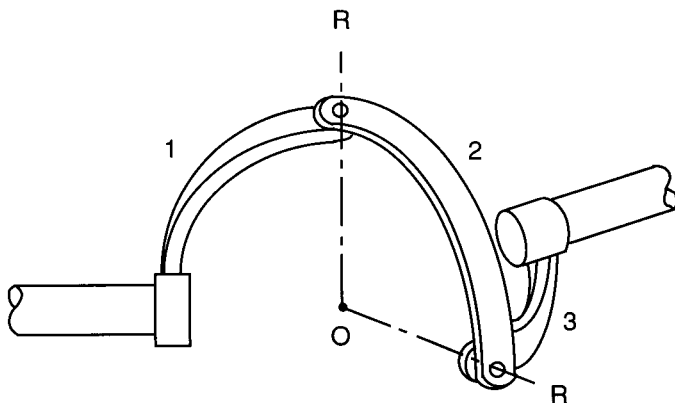


FIGURE 1.4. Universal joint.

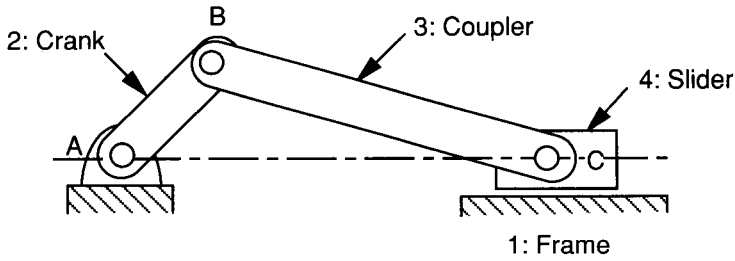


FIGURE 1.5. Slider-and-crank mechanism.

A kinematic chain is called a *mechanism* when one of its links is fixed to the ground. The fixed link is sometimes called the *base*. In a mechanism, one or more links may be assigned as the input links. As the input link(s) move with respect to the fixed link, all the other links will move according to the kinematic constraints imposed by the joints. Thus a mechanism is a device that transforms motion or torque from one or more input links to the others. Figure 1.5 shows a slider-and-crank mechanism that can transform the crank rotation into a reciprocating motion of the slider, and vice versa.

A *machine* is an assemblage of one or more mechanisms, along with other electrical and/or hydraulic components, used to transform external energy into useful work or other form of energy. For example, an electric drill is a machine. It consists of an electric motor, a speed reduction unit, a keyed chuck, and a trigger switch for the purpose of transforming electric energy into drilling work. However, the speed reduction unit itself is a mechanism and not a machine.

The terms *mechanism* and *machine* are sometimes used synonymously. According to the definitions above, however, there is a definite difference. An assemblage of parts is called a *mechanism* if it is used only for the transmission of motion, and it is called a *machine* if it is used to transform external energy into useful work. The mechanical manipulator of a robotic system is a mechanism. For the mechanism to become a machine, a microprocessor-based controller, encoders and/or force sensors, and other accessories, such as a computer vision system, must be incorporated so that an external source of energy can be converted into useful work. Although a machine may consist of one or more mechanisms, a mechanism is not necessarily a machine since it does not necessarily have to do work but serves merely as a motion transformation device.

1.2.3 Degrees of Freedom of a Mechanism

Perhaps the first concern in a study of the kinematics of mechanisms is the number of *degrees of freedom*. The degrees of freedom of a mechanism are

the number of independent parameters or inputs needed to specify the configuration of the mechanism completely. Except for some special cases, it is possible to derive a general expression for the degrees of freedom of a mechanism in terms of the number of links, number of joints, and types of joints incorporated in the mechanism. The following notations are defined to facilitate the derivation of an equation:

c_i : number of constraints imposed by joint i .

F : degrees of freedom of a mechanism.

f_i : degrees of relative motion permitted by joint i .

j : number of joints in a mechanism, assuming that all the joints are binary.

j_i : number of joints with i degrees of freedom.

L : number of independent loops in a mechanism.

n : number of links in a mechanism, including the fixed link.

λ : degrees of freedom of the space in which a mechanism is intended to function.

Since we assume that all the joints are binary, a ternary joint is counted as two binary joints, a quaternary joint is counted as three binary joints, and so on. We also assume that a single value of λ applies to the motions of all the moving links. That is, they all operate in the same working space: hence $\lambda = 6$ for spatial mechanisms, and $\lambda = 3$ for planar and spherical mechanisms.

Intuitively, the degree-of-freedom value of a mechanism is equal to the degrees of freedom associated with all the moving links minus the number of constraints imposed by the joints. Hence, if the links are all free of constraints, the degrees of freedom of an n -link mechanism, with one of its links fixed to the ground, would be equal to $\lambda(n - 1)$. However, the total number of constraints imposed by the joints is equal to $\sum_{i=1}^j c_i$. Hence the degree-of-freedom value of a mechanism is generally given by

$$F = \lambda(n - 1) - \sum_{i=1}^j c_i. \quad (1.1)$$

The number of constraints imposed by a joint and the degrees of freedom permitted by the joint are equal to the motion parameter, λ ; that is,

$$\lambda = c_i + f_i. \quad (1.2)$$

Hence the total number of constraints imposed by the joints is

$$\sum_{i=1}^j c_i = \sum_{i=1}^j (\lambda - f_i) = j\lambda - \sum_{i=1}^j f_i. \quad (1.3)$$

Substituting Eq. (1.3) into (1.1) yields

$$F = \lambda(n - j - 1) + \sum_i f_i. \quad (1.4)$$

Equation (1.4) is known as the *Grübler* (1917) or *Kutzbach* (1929) *criterion*. In reality, the criterion was established much earlier by *Ball* (1900) and probably by others. However, unlike earlier researchers, *Grübler* and *Kutzbach* developed the equation specifically for mechanisms.

Example 1.2.1 Four-Bar Linkage Figure 1.6 shows a planar four-bar linkage. There are four links connected in a loop by four revolute joints. Hence $\lambda = 3$, $n = 4$, $j = j_1 = 4$, and Eq. (1.4) yields $F = 3(4 - 4 - 1) + 4 \times 1 = 1$. The planar four-bar linkage is often employed in a manipulator as a push-rod linkage for transmitting the rotation of a base- or near-base-connected actuator to the rotation of a link at the far end of a manipulator, as shown in Fig. 1.2, links 2–3–8–7.

Example 1.2.2 Five-Bar Linkage Figure 1.7 shows a planar five-bar linkage in which all the joints are revolute. We have $\lambda = 3$, $n = 5$, and $j = j_1 = 5$. Hence Eq. (1.4) yields $F = 3(5 - 5 - 1) + 5 \times 1 = 2$. The planar five-linkage is often used as a 2-dof manipulator. Using 1 and 2 as the input links, point Q can be positioned anywhere on the plane.

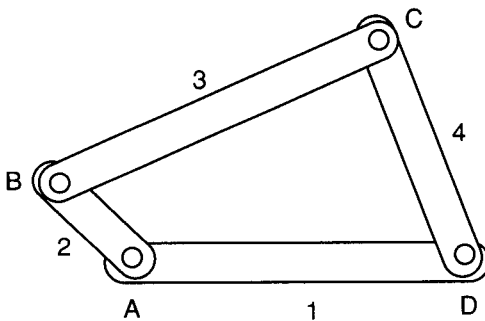


FIGURE 1.6. Four-bar linkage.

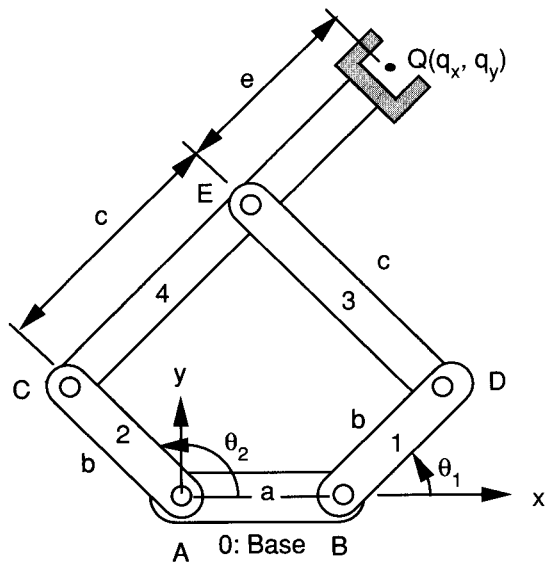


FIGURE 1.7. Five-bar, 5R manipulator.

Example 1.2.3 Pantograph Mechanism Figure 1.8 shows a planar pantograph mechanism in which the two ground-connected joints are prismatic and the remaining joints are revolute. For this mechanism we have $\lambda = 3$, $n = 7$ (including the ground link), and $j = j_1 = 8$. Equation (1.4) yields $F = 3(7 - 8 - 1) + 8 \times 1 = 2$. Hence the pantograph mechanism is often employed as a 2-dof manipulator. Using 1 and 2 as the input links, point Q can be positioned anywhere on the plane.

The Grübler criterion is valid provided that the constraints imposed by the joints are independent of one another and do not introduce redundancies. For example, a revolute–spherical binary link chain with the revolute joint axis passing through the center of the spherical joint introduces a redundant degree of freedom. This type of freedom, called a *passive degree of freedom*, allows an intermediate link to rotate freely about an axis defined by the two joints. Although the intermediate link does have the ability to transmit forces or torques and therefore motion about some other axes, it has no torque transmission capability about the passive axis. In general, binary links with S – S , S – E , and E – E pairs possess a passive degree of freedom (Table 1.1). A sequence of binary links with S – S , S – E , and E – E pairs as their terminal joints also possesses a passive degree of freedom.

A passive degree of freedom cannot transmit torque, and therefore motion, about the passive axis. When such a joint pair exists in a mechanism, 1 degree

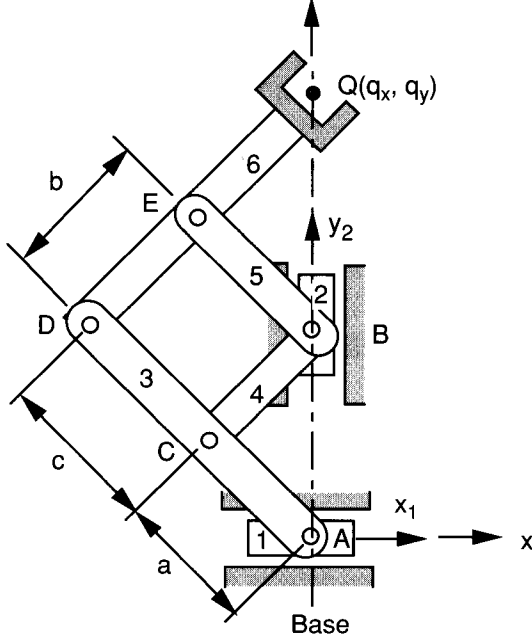


FIGURE 1.8. Pantograph mechanism.

of freedom should be subtracted from the degree-of-freedom equation. Let f_p be the number of passive degrees of freedom in a mechanism; then the remaining active degrees of freedom is given by

$$F = \lambda(n - j - 1) + \sum_i f_i - f_p. \quad (1.5)$$

Example 1.2.4 Stewart–Gough Platform Figure 1.9 shows a spatial mechanism in which a moving platform is connected to a fixed base with six extensible limbs by spherical joints. Each limb is made up of two

TABLE 1.1. Binary Links with Passive Degrees of Freedom

Type	Passive Degrees of Freedom
$S-S$	Rotation about an axis through centers of ball joints.
$S-E$	Rotation about an axis through the center of the ball and perpendicular to the plane of the plane pair.
$E-E$	Sliding along an axis parallel to the line of intersection of the planes of the plane pairs. If the two planes are parallel, 3 passive degrees of freedom exist.

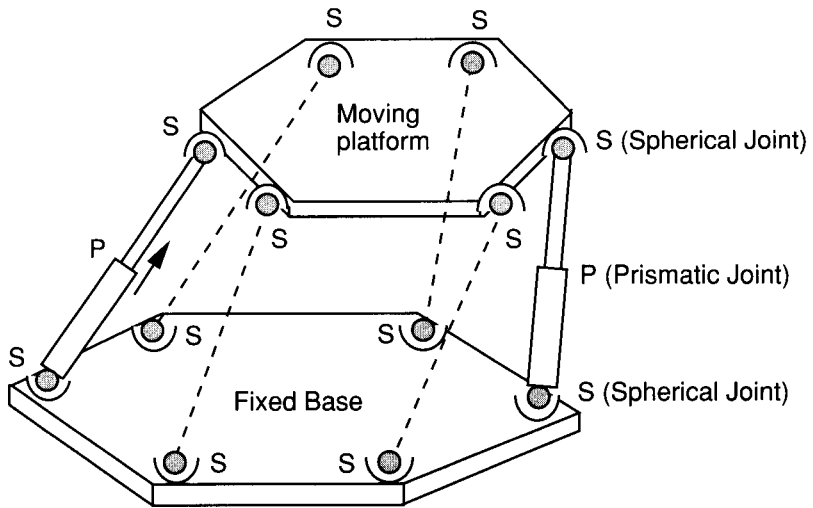


FIGURE 1.9. Stewart-Gough platform.

binary links that are connected by a prismatic joint. This construction is called an S - P - S limb. Because of the S - P - S combination, there is a passive degree of freedom associated with each limb. Hence we have $\lambda = 6$, $n = 14$, $j_1 = 6$, $j_3 = 12$, and $f_p = 6$. The degree-of-freedom value of the manipulator is given by $F = 6(14 - 18 - 1) + (12 \times 3 + 6) - 6 = 6$.

Since Eq. (1.5) is derived by using a single value of λ for all the moving links, the degree of freedom of a spatial mechanism with a planar or spherical linkage as a subsystem should be counted with care. Specifically, $\lambda = 3$ should be used for the planar or spherical subsystem, while $\lambda = 6$ is used for the spatial portion of the mechanism. For example, the manipulator shown in Fig. 1.2 consists of a planar four-bar linkage (links 2, 3, 8, and 7) and a spherical wrist mechanism (links 3, 4, 5, and 6) as two subsystems. The four-bar linkage provides 1 degree of freedom, the spherical linkage gives 3 degrees of freedom, and the open-loop chain 0-1-2 provides 2 additional degrees of freedom. Overall, it is a 6-dof spatial manipulator.

In general, if the Grübler criterion yields $F > 0$, the mechanism has F degrees of freedom. If the criterion yields $F = 0$, the mechanism becomes a structure with zero degrees of freedom. On the other hand, if the criterion yields $F < 0$, the mechanism becomes a statically indeterminate structure with redundant constraints. However, it should be noted that there are mechanisms that do not obey the Grübler criterion. Those mechanisms that require special link lengths to achieve mobility are called *overconstrained mechanisms* (Mavroidis and Roth, 1994).

For closed-loop mechanisms and mechanical manipulators, the number and locations of actuated joints should be chosen carefully such that the end effector can be controlled at will. In general, the number of actuated joints should be equal to the number of degrees of freedom of the mechanism, and the locations of actuated joints should be chosen such that they form a set of independent coordinates. If the number of actuators is less than the number of degrees of freedom, the output link cannot be controlled at will. On the other hand, if the number of actuators is greater than the number of degrees of freedom, the motion of these actuators must be coordinated in accordance with their kinematic constraints. A redundantly driven manipulator can be employed for the purpose of eliminating gear backlash and clearances associated with the geometry of the joints (Chang and Tsai, 1993; Tsai and Chang, 1994).

1.2.4 Loop Mobility Criterion

As stated earlier, a kinematic chain is an assemblage of links connected by joints. If every link is connected to at least two other links, the chain forms one or more closed loops. It is possible to establish an equation relating the number of independent loops to the number of links and joints in a kinematic chain. Here the term *number of independent loops* refers to the total number of loops, excluding the external loop.

We observe that the four-bar linkage shown in Fig. 1.6 is a kinematic chain with one independent loop. It is made up of four links and four joints. The five-bar linkage shown in Fig. 1.7 is also a kinematic chain with one independent loop, and it is made up of five links and five joints. We conclude that for a single-loop kinematic chain (planar, spherical, or spatial), the number of links is equal to the number of joints. That is, $n = j$ for all single-loop kinematic chains.

We now extend a single-loop chain into a two-loop chain. This can be accomplished by taking an open-loop chain and joining its two ends to any two members of a single-loop chain by two joints, as depicted in Fig. 1.10, where A and B are the attachment points. Note that by extending from one to two loops, the number of joints added exceeds the number of links by one. Similarly, an open-loop chain can be added to a two-loop chain to form a three-loop chain, and so on. In general, extending a kinematic chain from one to L loops, the difference between the number of joints and number of links is increased by $L - 1$. Hence we can write

$$j = n + L - 1, \quad (1.6)$$

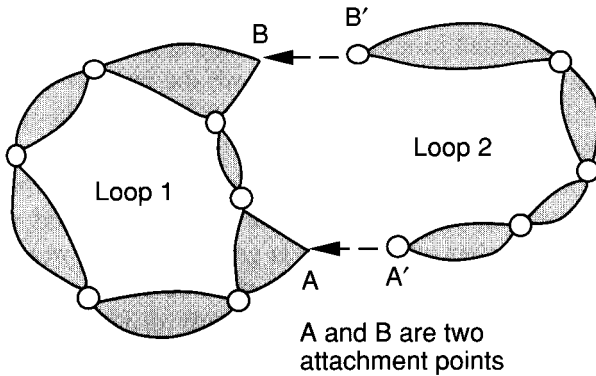


FIGURE 1.10. Formation of a multiloop chain.

or

$$L = j - n + 1. \quad (1.7)$$

Equation (1.7) is known as *Euler's equation*. It states that the number of independent loops exceeds the difference between the number of joints and the number of links by one. Combining Eq. (1.7) with (1.4) yields

$$\sum f_i = F + \lambda L. \quad (1.8)$$

Equation (1.8) is known as the *loop mobility criterion*. Equation (1.7) or (1.8) is useful for predicting the number of independent loops in a kinematic chain. Generally, the number of *loop-closure equations* that can be established for a mechanism is equal to the number of independent loops in the mechanism. This is demonstrated in Chapters 3 and 5.

Example 1.2.5 Pantograph Mechanism For the planar pantograph mechanism shown in Fig. 1.8, we have $\lambda = 3$, $n = 7$, and $j = 8$. Hence there are $L = 8 - 7 + 1 = 2$ independent loops. The first loop consists of the base link and links 1, 3, 4, and 2. The second loop consists of links 3, 4, 5, and 6.

Example 1.2.6 Stewart–Gough Platform For the Stewart–Gough platform shown in Fig. 1.9, we have $\lambda = 6$, $n = 14$, and $j = 18$. Hence there are $L = 18 - 14 + 1 = 5$ independent loops. The right-hand side of Eq. (1.8) yields $F + \lambda L = 6 + 6 \times 5 = 36$, while the left-hand side gives $\sum f_i = 3 \times 12 + 6 = 42$. The difference is due to the existence of the 6 passive degrees of freedom.

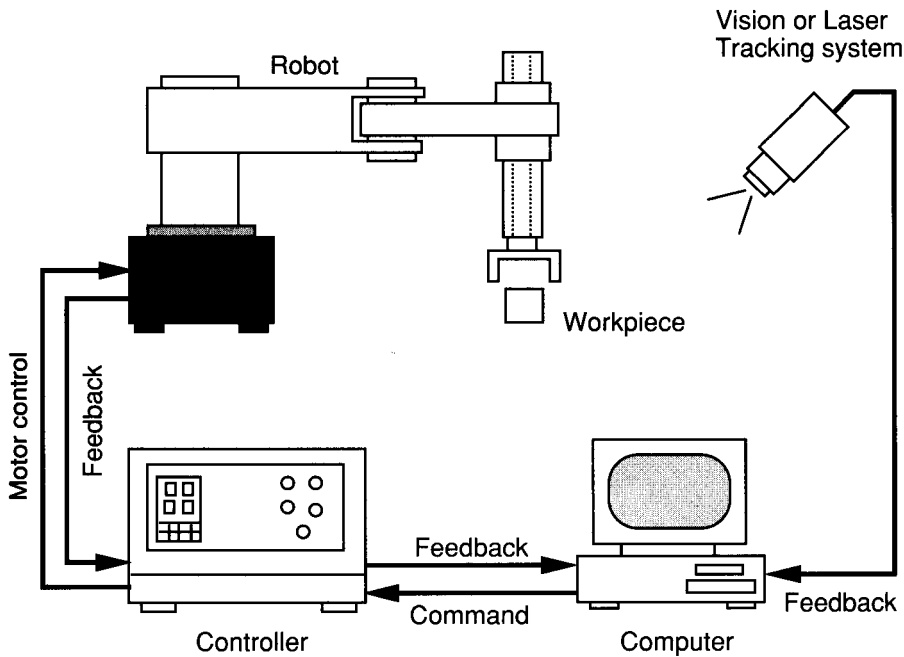


FIGURE 1.11. Typical robotic system.

1.3 ROBOTIC SYSTEMS

A robotic system typically consists of a mechanical manipulator, an end effector, a microprocessor-based controller, a computer, and perhaps, a vision system or other noncontact sensing devices (Fig. 1.11).

A *mechanical manipulator* is made up of several links connected by joints. One link is fixed to the ground, while another is designated as the *output link*. Some of the joints in the manipulator are actuated; the others are passive. Typically, the number of actuated joints is equal to the number of degrees of freedom so that the manipulator can be controlled at will. The manipulator shown in Fig. 1.2 is a 6-dof manipulator in which link 0 is fixed to the ground and link 6 is the output link. The joints between links (0 and 1), (1 and 2), (1 and 7), (3 and 4), (4 and 5), and (5 and 6) are actuated; the others are passive.

End effectors are devices attached at the output link of a mechanical manipulator to grasp, lift, and manipulate workpieces. We may think of the end effector as a mechanical interface between a mechanical manipulator and its environment. Generally, an end effector is a specialized device that is used for handling a few objects of similar shape, size, and weight in a repetitive operation where the requirement for dexterity and versatility is not essential. Such

an end effector is sometimes called a *gripper*. Most simple grippers are 1-dof devices. Figure 1.12 shows a rack-and-pinion gripper driven by a pneumatic source, and Fig. 1.13 shows a gripper equipped with force sensor.

For some applications, an end effector may be required to handle a variety of objects of different shapes, sizes, and materials. We call an end effector that requires certain dexterity and versatility a *universal gripper* or *dextrous hand*. The Stanford/JPL hand (Salisbury, 1982; Salisbury and Roth, 1983) and the Utah/MIT hand (Jacobsen et al., 1984, 1986) are dextrous hands. Figure 1.14 shows a Utah/MIT hand and the mechanical components and some sensing elements of the hand are shown in Fig. 1.15. The basics of gripping and a variety of grippers can be found in Pham and Heginbotham (1986).

The *controller* may range from a simple low-level PID controller to a high-level model-based intelligent controller. Encoders and/or tachometers are installed at the joints of a manipulator to measure the relative motion between two connected bodies for the purpose of feedback control. Some advanced intelligent controllers may require the manipulator to be equipped with force and/or touch sensors along with a vision system or some other sensor. In addition, the controller is often equipped with a *teach pendant* that is capable of teaching position, editing and running programs, and interfacing with the computer.

The computer may be microprocessor-based hardware, a personal computer, or a mainframe computer capable of high-speed real-time computation, reasoning, and issuing intelligent control commands.

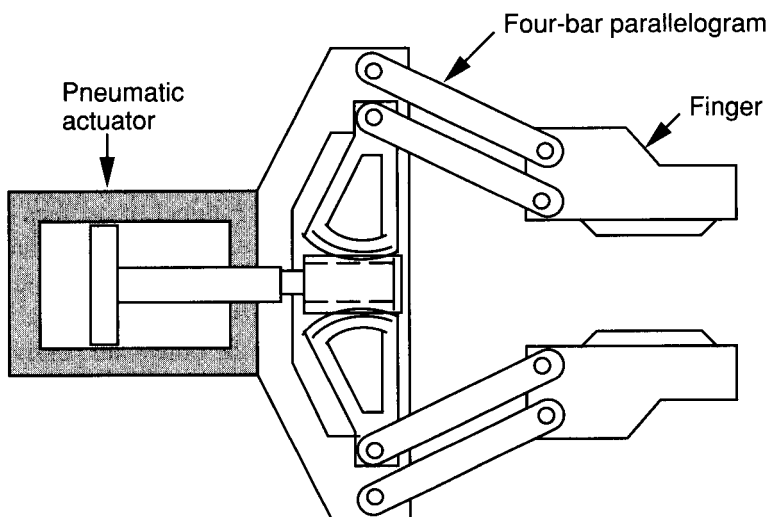


FIGURE 1.12. Dual rack-and-pinion gripper mechanism.

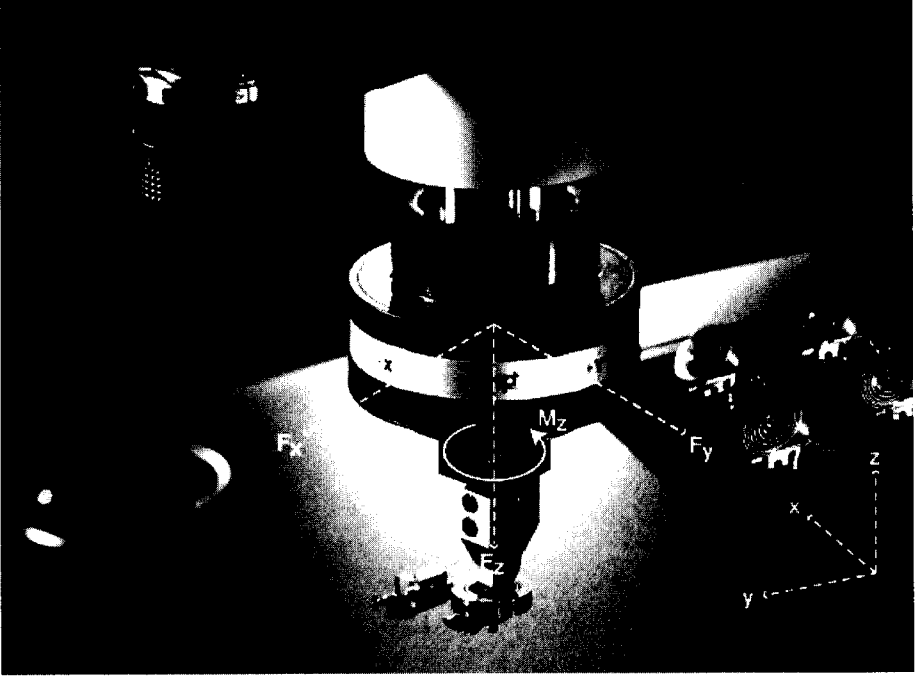


FIGURE 1.13. Adept gripper. (Courtesy of Adept Technology, Inc., San Jose, California.)

1.4 CLASSIFICATION OF ROBOTS

Robots can be classified according to various criteria, such as their degrees of freedom, kinematic structure, drive technology, workspace geometry, and motion characteristics. We discuss each of these in turn.

1.4.1 Classification by Degrees of Freedom

One obvious classification scheme is to categorize robots according to their degrees of freedom. Ideally, a manipulator should possess 6 degrees of freedom in order to manipulate an object freely in three-dimensional space. From this point of view, we call a robot a *general-purpose robot* if it possesses 6 degrees of freedom, a *redundant robot* if it possesses more than 6 degrees of freedom, and a *deficient robot* if it possesses less than 6 degrees of freedom. A redundant robot provides more freedom to move around obstacles and operate in a tightly confined workspace. On the other hand, for some special applications, such as assembling components on a plane, a robot with just

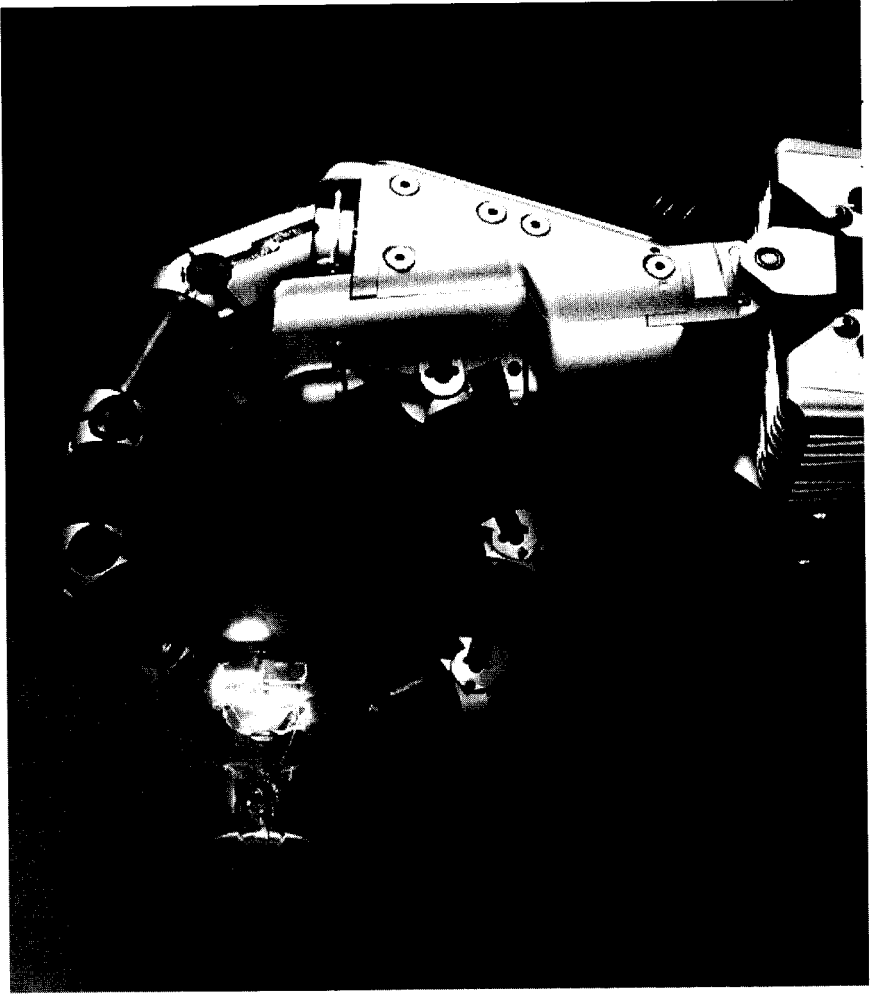


FIGURE 1.14. Utah/MIT dextrous hand. (Courtesy of SARCOS, Salt Lake City, Utah.)

4 degrees of freedom would be sufficient. The Fanuc S-900W robot shown in Fig. 1.2 is a 6-dof general-purpose manipulator and the Adept-One robot shown in Fig. 1.16 is a 4-dof manipulator.

1.4.2 Classification by Kinematic Structure

Another scheme is to classify robots according to their structural topologies. A robot is said to be a *serial robot* or *open-loop manipulator* if its kinematic structure takes the form of an open-loop chain, a *parallel manipulator* if it is



FIGURE 1.15. Components of the Utah/MIT hand. (Courtesy of SARCOS, Salt Lake City, Utah.)

made up of a closed-loop chain, and a *hybrid manipulator* if it consists of both open- and closed-loop chains. The Adept-One robot shown in Fig. 1.16 is a serial manipulator, and the manipulator shown in Fig. 1.17 is a parallel manipulator. Most engineers consider the Fanuc S-900W robot shown in Fig. 1.2 to be a serial manipulator. In reality, it is a hybrid manipulator, since it incorporates a four-bar (push-rod) linkage to drive the third joint. Many industrial robots employ this type of construction because it allows the third motor to be mounted on the waist and therefore reduces the inertia of the manipulator. In general, a parallel manipulator has the advantages of higher stiffness, higher payload capacity, and lower inertia to the manipulation problem than a comparable serial manipulator, at the price of a smaller workspace and more complex mechanism.

1.4.3 Classification by Drive Technology

Manipulators can also be classified by their drive technology. The three popular drive technologies are *electric*, *hydraulic*, and *pneumatic*. Most manip-

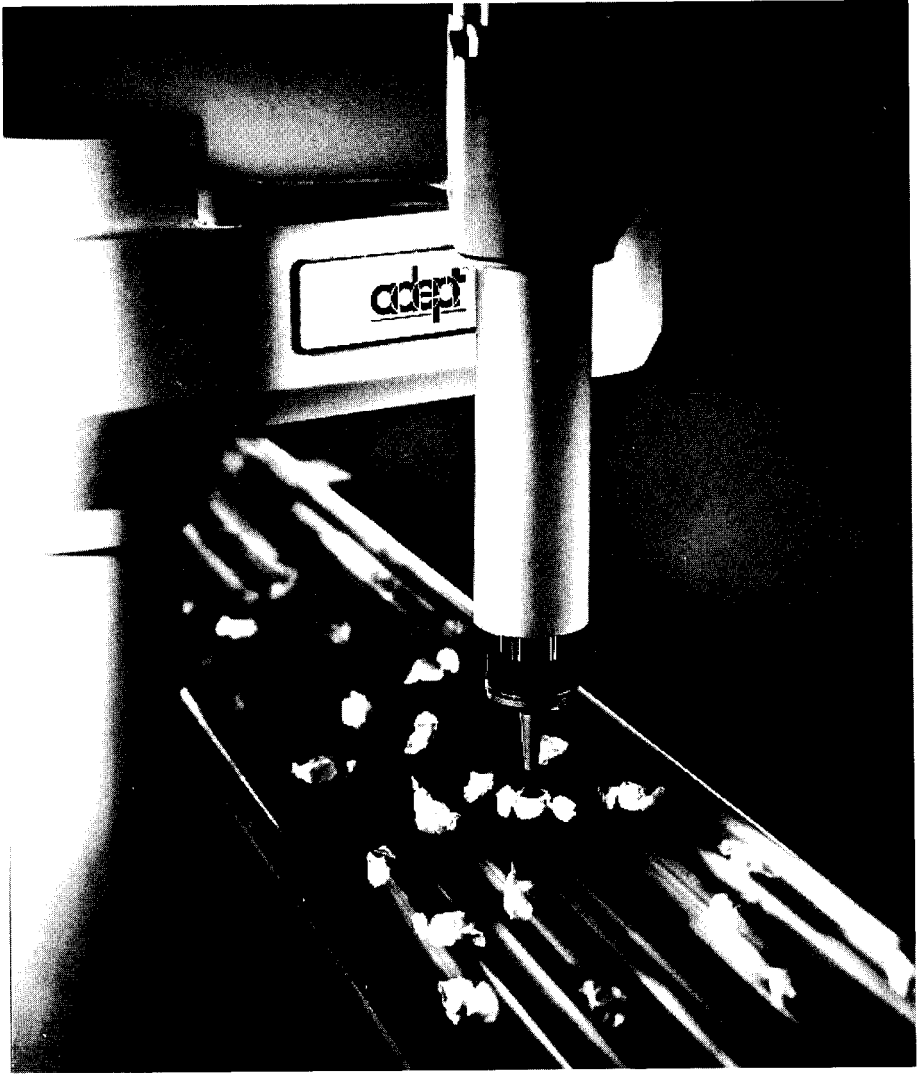


FIGURE 1.16. Adept-One robot. (Courtesy of Adept Technology, Inc., San Jose, California.)

ulators use either electric dc servomotors or stepper motors, because they are clean and relatively easy to control. However, when high-speed and/or high-load-carrying capabilities are needed, hydraulic or pneumatic drive is preferred. A major disadvantage associated with the use of a hydraulic drive is the possibility of leaking oils. Additionally, a hydraulic drive is inherently flexible, due to the bulk modulus of oil. Although a pneumatic drive is clean and fast, it is difficult to control because air is a compressible fluid.



FIGURE 1.17. A 4-dof parallel robot. (Courtesy of Demarex Robotique & Microtechnique S.A., Switzerland.)

In a serial manipulator, typically, one actuator is used to control the motion of each joint. If every moving link is driven by one actuator mounted on its preceding link through a gear reduction unit, the joint displacements are kinematically independent of one another. We call this type of manipulator a *conventional serial manipulator*. On the other hand, if each joint is driven directly by an actuator without a speed reducer, the manipulator is called a *direct-drive manipulator*.

Employing a gear reduction unit permits the use of a smaller motor, which in turn reduces the inertia of a manipulator. However, backlash in a gear reduction unit can cause position error at the end effector. Direct-drive technique eliminates the problems associated with gear backlash and can possibly increase the speed of a manipulator. However, direct-drive motors are bulky and heavy. Consequently, they are commonly used to drive the first joint of

a manipulator where the motor can be installed on the base. Alternatively, they can also be installed on the base to drive the second or third joint via a push-rod linkage or a metal belt. The first joint of the Adept-One robot shown in Fig. 1.16 is directly driven by a direct-drive motor, and the second joint is driven by a direct-drive motor via two pretensioned opposing metal belts.

Some manipulators employ gear trains, tendons, or chain-and-sprocket devices to drive their joints. When such transmission mechanisms are used to drive a manipulator across several joints, the joint displacements are no longer independent of each other. Strictly speaking, such manipulators should be considered as closed-loop manipulators, although there may be several articulation points. Figure 1.18 shows a 5-dof educational robot. The mechanical transmission system of this robot consists of gears, timing belts, and lead screws.

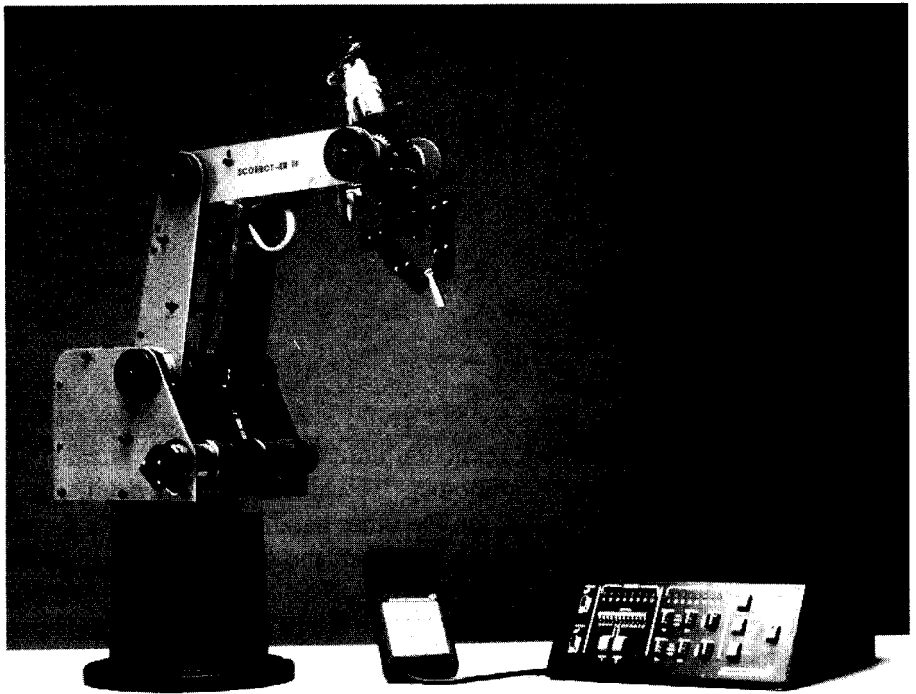


FIGURE 1.18. Scorbot-ER III robot. (Courtesy of Eshed Robotec Inc., Princeton, New Jersey.)

1.4.4 Classification by Workspace Geometry

The *workspace* of a manipulator is defined as the volume of space the end effector can reach. Two different definitions of workspace are frequently used. A *reachable workspace* is the volume of space within which every point can be reached by the end effector in at least one orientation. A *dextrous workspace* is the volume of space within which every point can be reached by the end effector in all possible orientations. Clearly, the dextrous workspace is a subset of the reachable workspace.

Although it is not a necessary condition, many serial manipulators are designed with their first three moving links longer than the remaining links. Thus the first three links are used primarily for manipulating the position and the remaining links for controlling the orientation of the end effector. For this reason, the subassembly associated with the first three links is called the *arm*, and the subassembly associated with the remaining links is called the *wrist*. Except for redundant manipulators, the arm usually possesses 3 degrees of freedom while the wrist may have 1 to 3 degrees of freedom. Furthermore, the wrist assembly is often designed with its joint axes intersecting at a common point called the *wrist center*. The arm assembly can assume various kinematic structures and therefore generates different work envelopes called *regional workspaces*. The workspace supplied by robotic manufacturers usually shows the regional workspace.

Perhaps the simplest kinematic structure of a robot arm is made up of three mutually perpendicular *prismatic joints*. This type of robot is known as a *Cartesian robot*. The wrist center position of a Cartesian robot can be conveniently described by three Cartesian coordinates associated with the three prismatic joints. Obviously, the regional workspace of a Cartesian robot is a rectangular box. Figure 1.19 shows a Cartesian robot manufactured by Seiko Instruments. When a Cartesian robot is mounted on rails above its workspace, it is called a *gantry robot*.

A robot arm is called a *cylindrical robot* if either the first or second joint of a Cartesian robot is replaced by a *revolute joint*. The wrist center position of a cylindrical robot can be described by a set of cylindrical coordinates associated with the three joint variables. The prismatic joints usually have mechanical limits on both ends. Hence the workspace of a cylindrical robot is confined by two concentric cylinders of finite length. Figure 1.20 shows a cylindrical robot.

A robot arm is called a *spherical robot* if the first two joints are made up of two intersecting revolute joints and the third is a prismatic joint. Normally, the prismatic joint is not parallel to the second joint axis. The wrist center position of a spherical robot can be described by a set of spherical coordinates

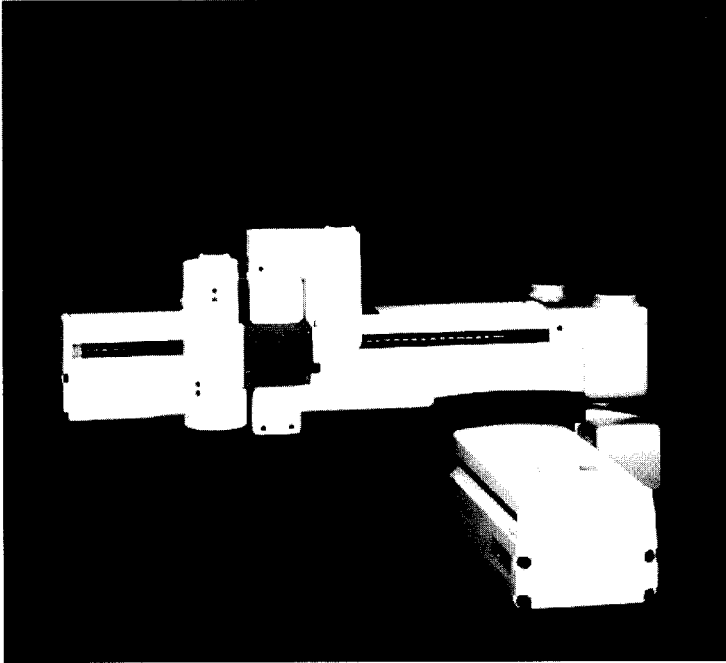


FIGURE 1.19. Cartesian robot. (Courtesy of Seiko Instruments USA, Inc., Torrance, California.)

associated with the three joint variables. Hence the workspace of a spherical robot is confined by two concentric spheres.

A robot arm is said to be an *articulated robot* if all three joints are revolute. The workspace of an articulated robot is very complex, typically a crescent-shaped cross section. Many industrial robots are of the articulated type. The PUMA robot is an articulated robot, as is the master-slave robot shown in Fig. 1.3.

The *SCARA* (selective compliance assembly robot arm) *robot* is a special type of robot. It consists of two revolute joints followed by a prismatic joint. In addition, all three joint axes are parallel to each other and usually point along the direction of gravity. Thus the first two actuators do not have to work against the gravitational forces of the links and the payload. The wrist usually has only 1 degree of freedom. Hence the entire robot has 4 degrees of freedom. This type of manipulators is useful for assembling parts on a plane. The Adept robot shown in Fig. 1.16 is a SCARA robot.

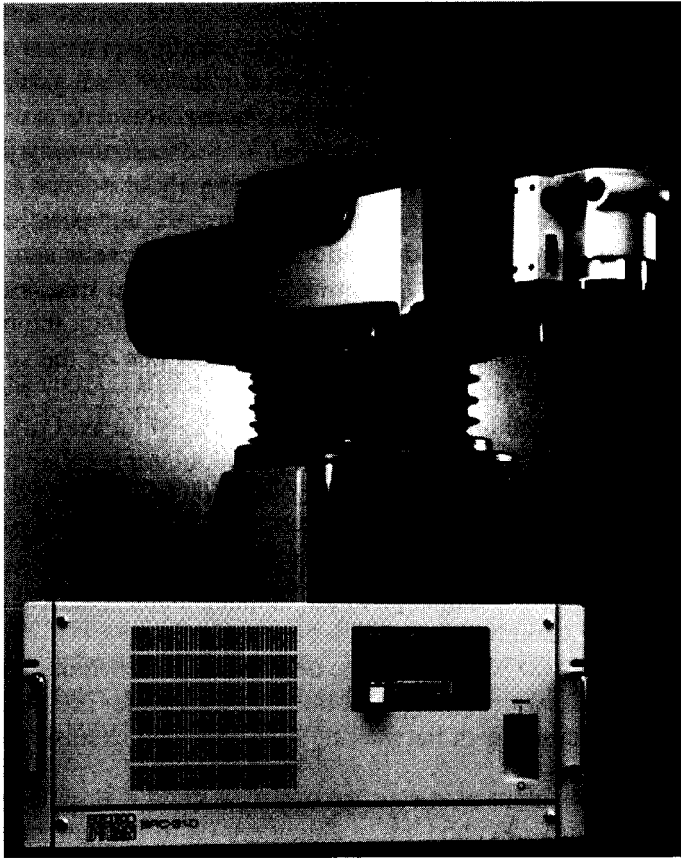


FIGURE 1.20. Cylindrical robot. (Courtesy of Seiko Instruments USA, Inc., Torrance, California.)

1.4.5 Classification by Motion Characteristics

Robot manipulators can also be classified according to their nature of motion. A rigid body is said to perform a *planar motion* if all particles in the body describe plane curves that lie in parallel planes. A mechanism is said to be a *planar mechanism* if all the moving links in the mechanism perform planar motions that are parallel to one another. For a planar mechanism, the loci of all points in all links can be drawn conveniently on a plane. Planar mechanisms that utilize only lower-pair joints are called *planar linkages*. Revolute and prismatic joints are the only permissible lower pairs for planar linkages. In a planar linkage, the axes of all revolute joints must be normal to the plane of motion, while the direction of translation of a prismatic joint must be parallel to the plane of motion. A manipulator is called a *planar manipulator* if its

mechanism is a planar mechanism, as are the manipulators shown in Figs. 1.7 and 1.8. Planar manipulators are useful for manipulating an object on a plane.

A rigid body is said to be under a *spherical motion* if all particles in the body describe curves that lie on concentric spheres. Thus when a rigid body performs a spherical motion, there exists at least one stationary point. Following the definition above, a rigid body rotating about a fixed axis can be considered as a special case of spherical motion since any point on the axis of revolution can be treated as the stationary point. A mechanism is said to be a *spherical mechanism* if all the moving links perform spherical motions about a common stationary point. In a spherical mechanism, the motions of all particles can be described by their radial projections on the surface of a unit sphere. A revolute joint is the only permissible lower-pair joint for the construction of spherical linkages. In addition, all the joint axes of a spherical linkage must intersect at a common point. A manipulator is called a *spherical manipulator* if it is made up of a spherical mechanism. Figure 1.21 shows a 3-dof spherical parallel manipulator. In this mechanism, all the revolute joints intersect at a common center point O . Links 1, 2, and 3 are considered as the input links, and link 7 is designated as the output link. A spherical manipulator can be used as a pointing device.

A rigid body is said to perform a *spatial motion* if its motion cannot be characterized as planar or spherical motion. A manipulator is called a *spatial manipulator* if at least one of the moving links in the mechanism possesses a

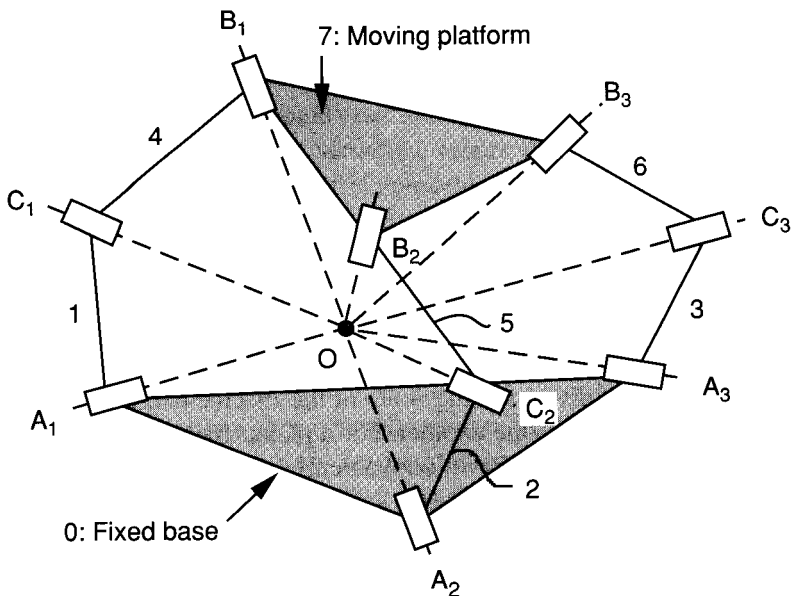


FIGURE 1.21. Spherical 3RRR manipulator.

general spatial motion. In general, we cannot associate unique motion characteristics with a spatial mechanism. The mechanism shown in Fig. 1.9 is a spatial 6-dof parallel manipulator. Planar and spherical mechanisms can be considered as special cases of spatial mechanisms. They occur as a consequence of special geometry in the particular orientations of their joint axes. Because of the special geometric conditions, the synthesis and analysis problems associated with planar and spherical mechanisms are greatly simplified. The selection of a robot manipulator depends on the application, the working environment, and other considerations.

1.5 POSITION, ORIENTATION, AND LOCATION OF A RIGID BODY

In the study of the kinematics of robot manipulators, we are constantly dealing with the *location* of several bodies in space. The bodies of interest include the links of a manipulator, the tools, the workpieces, and so on. To identify the location of a body, a reference coordinate system is established. We called this reference coordinate system the *fixed frame*, although in reality, it may not necessarily be fixed to the ground. In what follows we employ a Cartesian coordinate system to describe the location of a body, although other types of coordinate systems, such as the *cylindrical coordinate system* and *spherical coordinate system*, may also be used.

The *location* of a body with respect to a reference coordinate system is known if the *position* of all the points of the body are known. If the body of interest is rigid, six independent parameters would be sufficient to describe its location in three-dimensional space. As shown in Fig. 1.22, we take the (x, y, z) coordinate system as the fixed frame. We also attach an (u, v, w) Cartesian coordinate system to the moving body and refer to it as the *moving frame*. Clearly, the positions of all the points of the rigid body can be determined when the location of the moving frame with respect to the fixed frame is known. This relative *location* can be considered as composed of the *position* of a point, say the origin Q , and the *orientation* of the moving frame with respect to the fixed frame. Further, if we assume that the moving frame coincides with the fixed frame initially, the location of the moving frame with respect to the fixed frame and the spatial displacement of a rigid body from the initial coincident position are equivalent.

1.5.1 Description of a Position

The *position* of any point with respect to the reference frame can be described by a 3×1 *position vector*. For example, the position of a point P in the

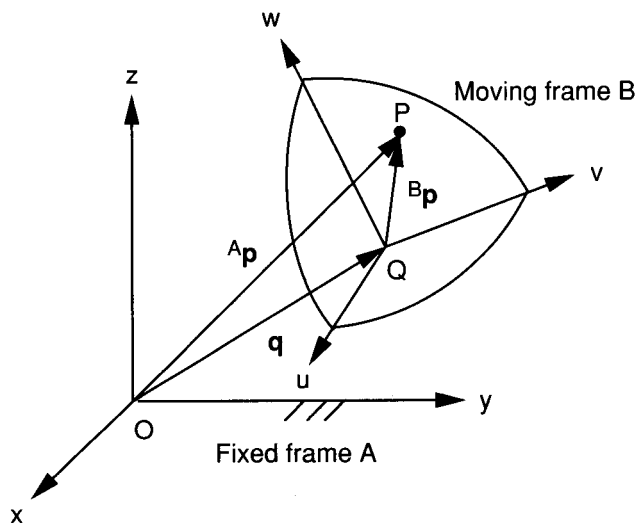


FIGURE 1.22. General spatial displacement.

reference frame A as shown in Fig. 1.23 is written as

$${}^A\mathbf{p} = \begin{bmatrix} p_x \\ p_y \\ p_z \end{bmatrix}, \quad (1.9)$$

where the subscripts x , y , and z represent the projections of the position vector onto the three coordinate axes of the reference frame. Since we are dealing

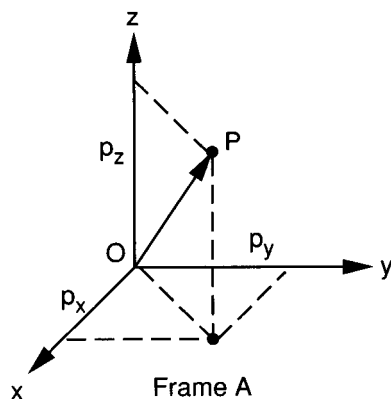


FIGURE 1.23. Position vector of a point P in space.

with several coordinate systems, a leading superscript is used to indicate the coordinate system to which the vector is referred.

1.5.2 Description of an Orientation

The *orientation* of a rigid body with respect to the fixed frame can be described in several different ways. In what follows we first describe the *direction cosine representation* followed by the *screw axis representation* and then the *Euler angle representation*. To describe the orientation of a rigid body, we consider the motion of a moving frame B with respect to a fixed frame A with one point fixed. This is known as a *rotation* or a *spherical motion*. Without losing generality, we assume that the origin of the moving frame is fixed to that of the fixed frame, as shown in Fig. 1.24.

(a) Direction Cosine Representation. One convenient way of describing the orientation of a rigid body is by means of the *direction cosines* of the coordinate axes of the moving frame with respect to the fixed frame. Let $\mathbf{i}$, $\mathbf{j}$, and $\mathbf{k}$ denote three unit vectors pointing along the coordinate axes of the fixed frame A , and $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ denote three unit vectors pointing along the coordinate axes of the moving frame B , respectively, as shown in Fig. 1.24. The three unit vectors $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ can be expressed in the fixed frame A as follows:

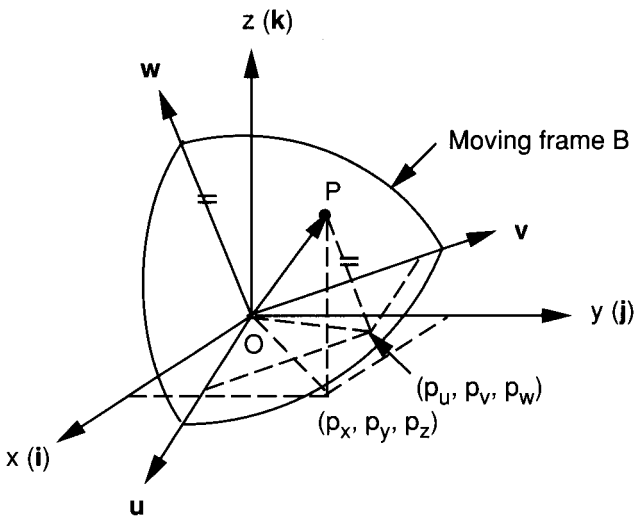


FIGURE 1.24. Spherical displacement.

$$\begin{aligned}
{}^A\mathbf{u} &= u_x\mathbf{i} + u_y\mathbf{j} + u_z\mathbf{k}, \\
{}^A\mathbf{v} &= v_x\mathbf{i} + v_y\mathbf{j} + v_z\mathbf{k}, \\
{}^A\mathbf{w} &= w_x\mathbf{i} + w_y\mathbf{j} + w_z\mathbf{k}.
\end{aligned} \tag{1.10}$$

The position vector of a point P of the rigid body can be expressed either in the fixed frame A as

$${}^A\mathbf{p} = p_x\mathbf{i} + p_y\mathbf{j} + p_z\mathbf{k}, \tag{1.11}$$

or in the rotated frame B as

$${}^B\mathbf{p} = p_u\mathbf{u} + p_v\mathbf{v} + p_w\mathbf{w}. \tag{1.12}$$

Since P is a point of the rigid body, ${}^B\mathbf{p}$ is constant. However, ${}^A\mathbf{p}$ depends on the orientation of B relative to A . Substituting Eq. (1.10) in (1.12), we obtain the vector $\mathbf{p}$ expressed in the fixed frame A as

$$\begin{aligned}
{}^A\mathbf{p} &= (p_u u_x + p_v v_x + p_w w_x)\mathbf{i} + (p_u u_y + p_v v_y + p_w w_y)\mathbf{j} \\
&\quad + (p_u u_z + p_v v_z + p_w w_z)\mathbf{k}.
\end{aligned} \tag{1.13}$$

Equating the x , y , and z components of ${}^A\mathbf{p}$ in Eq. (1.13) to the corresponding components in (1.11) yields

$$\begin{aligned}
p_x &= u_x p_u + v_x p_v + w_x p_w, \\
p_y &= u_y p_u + v_y p_v + w_y p_w, \\
p_z &= u_z p_u + v_z p_v + w_z p_w.
\end{aligned} \tag{1.14}$$

Writing Eq. (1.14) in a matrix form we obtain

$${}^A\mathbf{p} = {}^A R_B {}^B\mathbf{p}, \tag{1.15}$$

where

$${}^A R_B = \begin{bmatrix} u_x & v_x & w_x \\ u_y & v_y & w_y \\ u_z & v_z & w_z \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}, \tag{1.16}$$

where the leading superscript A and the trailing subscript B indicate the order of transformation. In what follows we omit the leading superscript and trailing subscript from time to time whenever there are only two frames of reference and the order of transformation is clear.

We call the matrix ${}^A R_B$ the *rotation matrix* of the moving frame B with respect to the fixed frame A . The rotation matrix specifies the orientation of B completely with respect to A . It transforms the position vector of any point P from the moving frame B to the fixed frame A . From the definition above, we see that the columns of a rotation matrix represent three orthogonal unit vectors of the moving coordinate axes expressed in the fixed frame. Therefore, the rotation matrix is orthogonal. The orthogonality conditions can be stated as

$$\begin{aligned}\mathbf{u}^2 &= 1, \\ \mathbf{v}^2 &= 1, \\ \mathbf{w}^2 &= 1,\end{aligned}\tag{1.17}$$

and

$$\begin{aligned}\mathbf{u}^T \mathbf{v} &= 0, \\ \mathbf{v}^T \mathbf{w} &= 0, \\ \mathbf{w}^T \mathbf{u} &= 0.\end{aligned}\tag{1.18}$$

Because of the orthogonality conditions above, only three of the nine elements of ${}^A R_B$ are independent. Using the orthogonality conditions above, it can be shown that

$$\begin{aligned}\mathbf{u} \times \mathbf{v} &= \mathbf{w}, \\ \mathbf{v} \times \mathbf{w} &= \mathbf{u}, \\ \mathbf{w} \times \mathbf{u} &= \mathbf{v}.\end{aligned}\tag{1.19}$$

It can also be shown that

$$\det({}^A R_B) = 1.\tag{1.20}$$

Furthermore, due to the orthogonality conditions, the inverse transformation of a rotation matrix is equal to its transpose:

$${}^B R_A = {}^A R_B^{-1} = {}^A R_B^T.\tag{1.21}$$

Since the columns of ${}^A R_B$ represent three unit vectors of the coordinate axes of frame B expressed in frame A , it follows that the rows of ${}^A R_B$ represent the three unit vectors defined along the coordinate axes of frame A and expressed in frame B . Therefore, the rotation matrix can be interpreted as a

set of three column vectors or a set of three row vectors:

$${}^A R_B = \begin{bmatrix} {}^A \mathbf{u} & {}^A \mathbf{v} & {}^A \mathbf{w} \end{bmatrix} = \begin{bmatrix} {}^B \mathbf{i}^T \\ {}^B \mathbf{j}^T \\ {}^B \mathbf{k}^T \end{bmatrix}. \quad (1.22)$$

Euler's Theorem. Euler's theorem states that the general displacement of a rigid body with one point fixed is a *rotation* about some axis. This unique axis of rotation is called the *screw axis*. In what follows, we show how the screw axis can be derived from a given rotation matrix. 

Equation (1.15) provides an orthogonal transformation of the position of a point P in a moving frame B to a fixed frame A . Since the moving frame B coincides with the fixed frame A at the initial location, we may consider ${}^B \mathbf{p}$ as the first position and ${}^A \mathbf{p}$ as the second position of P of the rigid body B . Since the origin O is a stationary point, the screw axis passes through this point. Furthermore, if $\tilde{P}$ lies on the axis of rotation, its position vector will not be affected by the rotation; that is, 

$${}^B \tilde{\mathbf{p}} = {}^A \tilde{\mathbf{p}}. \quad (1.23)$$

The second position of P is governed by the orthogonal transformation given by Eq. (1.15). Substituting Eq. (1.23) into (1.15) and rearranging yields

$$({}^A R_B - I) {}^A \tilde{\mathbf{p}} = 0, \quad (1.24)$$

where I denotes a 3×3 identity matrix.

We note that Eq. (1.24) is a special case of the following general eigenvalue problem:

$$({}^A R_B - \lambda I) {}^A \tilde{\mathbf{p}} = 0, \quad (1.25)$$

where λ is called the *eigenvalue value* or *characteristic value*. Equation (1.25) consists of three linear homogeneous equations in three unknowns, $\tilde{p}_x$, $\tilde{p}_y$, and $\tilde{p}_z$. The compatibility condition for the existence of nontrivial solutions is that the determinant of the coefficients must vanish; that is, 

$$| {}^A R_B - \lambda I | = \begin{vmatrix} a_{11} - \lambda & a_{12} & a_{13} \\ a_{21} & a_{22} - \lambda & a_{23} \\ a_{31} & a_{32} & a_{33} - \lambda \end{vmatrix} = 0, \quad (1.26)$$

where a_{ij} represents the (i, j) element of ${}^A R_B$.

Equation (1.26) is known as the *characteristic equation* and the values of λ for which the equation is satisfied are the eigenvalues. In general, the characteristic equation will have three roots with three corresponding eigen-

vectors. Expanding Eq. (1.26), say along the first column of ${}^A R_B$ and applying Eqs. (1.19) and (1.20), we obtain

$$\lambda^3 - \text{tr}({}^A R_B)\lambda^2 + \text{tr}({}^A R_B)\lambda - 1 = 0, \quad (1.27)$$

where $\text{tr}({}^A R_B) = a_{11} + a_{22} + a_{33}$.

We note that Eq. (1.27) contains $\lambda - 1$ as a factor. Solving Eq. (1.27) yields

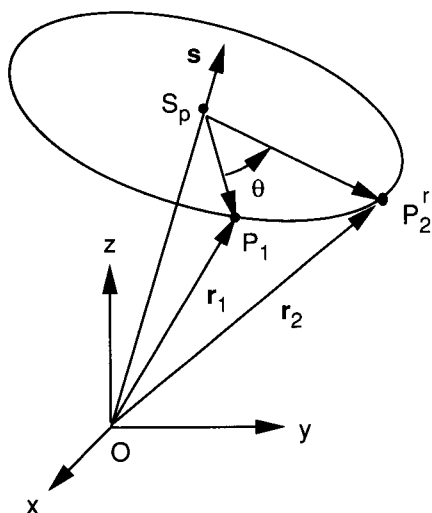
$$\lambda = 1, e^{i\theta}, e^{-i\theta},$$

where

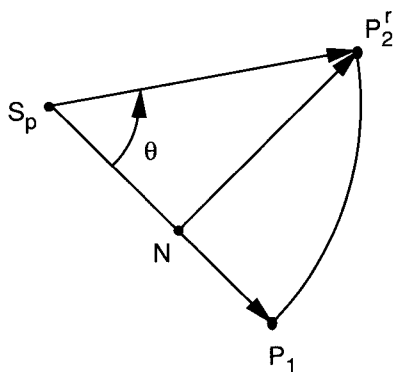
$$\theta = \cos^{-1} \frac{a_{11} + a_{22} + a_{33} - 1}{2} \quad (1.28)$$

defines the angle of rotation about the screw axis and where $\cos^{-1} x$ is the arccosine function. The eigenvector corresponding to the eigenvalue of $\lambda = 1$ gives the direction of the screw axis. Hence the direction of the screw axis is obtained by solving Eq. (1.24) for the ratio $\tilde{p}_x/\tilde{p}_y/\tilde{p}_z$.

(b) Screw Axis Representation. In this section we seek a description of the orientation of a rigid body in terms of a rotation about a screw axis. As shown in Fig. 1.25a, let the moving frame B be rotated an angle θ about an



(a) Spherical displacement



(b) Plane normal to the axis of rotation

FIGURE 1.25. Vector diagram of a spherical displacement.

axis passing through the origin of the fixed frame A . The first position of a point P of the rigid body B is denoted by the vector $\mathbf{r}_1 = \overline{OP}_1$. The second position is denoted by $\mathbf{r}_2 = \overline{OP}_2^r$, and the direction of rotation is denoted by a unit vector $\mathbf{s}(s_x, s_y, s_z)$. From the geometry of the figure, we obtain

$$\overline{S_p P_1} = \mathbf{r}_1 - (\mathbf{r}_1^T \mathbf{s})\mathbf{s}, \quad (1.29)$$

$$\overline{S_p P_2^r} = \mathbf{r}_2 - (\mathbf{r}_2^T \mathbf{s})\mathbf{s}. \quad (1.30)$$

Figure 1.25b shows a plane that contains both points P_1 and P_2^r and is normal to the axis of rotation. The point of intersection between the plane and the axis of rotation is taken as S_p . Let $\overline{N P_2^r} \perp \overline{S_p P_1}$. Then, using of the fact that $|\overline{S_p P_1}| = |\overline{S_p P_2^r}|$ and $\mathbf{s} \times \overline{S_p P_1} = \mathbf{s} \times \mathbf{r}_1$, we have

$$\overline{S_p N} = \overline{S_p P_1} c\theta, \quad (1.31)$$

$$\overline{N P_2^r} = \mathbf{s} \times \mathbf{r}_1 s\theta, \quad (1.32)$$

where $c\theta$ is a shorthand notation for $\cos \theta$ and $s\theta$ for $\sin \theta$. To derive a relation between $\mathbf{r}_2$ and $\mathbf{r}_1$, we express $\overline{S_p P_2^r}$ as a sum of two vectors:

$$\overline{S_p P_2^r} = \overline{S_p N} + \overline{N P_2^r}. \quad (1.33)$$

Substituting Eqs. (1.29) through (1.32) into (1.33), we obtain

$$\mathbf{r}_2 - (\mathbf{r}_2^T \mathbf{s})\mathbf{s} = [\mathbf{r}_1 - (\mathbf{r}_1^T \mathbf{s})\mathbf{s}]c\theta + \mathbf{s} \times \mathbf{r}_1 s\theta. \quad (1.34)$$

Substituting $\mathbf{r}_1^T \mathbf{s} = \mathbf{r}_2^T \mathbf{s}$ into Eq. (1.34) and rearranging yields

$$\mathbf{r}_2 = \mathbf{r}_1 c\theta + \mathbf{s} \times \mathbf{r}_1 s\theta + \mathbf{s}(\mathbf{r}_1^T \mathbf{s})(1 - c\theta). \quad (1.35)$$

Equation (1.35) is known as *Rodrigues's formula* for a spherical displacement of a rigid body. By considering $\mathbf{r}_1$ as ${}^B \mathbf{p}$ and $\mathbf{r}_2$ as ${}^A \mathbf{p}$, Eq. (1.35) can be written in matrix form as

$${}^A \mathbf{p} = {}^A R_B {}^B \mathbf{p}, \quad (1.36)$$

where the elements of the rotation matrix are given by:

$$a_{11} = (s_x^2 - 1)(1 - c\theta) + 1,$$

$$a_{12} = s_x s_y (1 - c\theta) - s_z s\theta,$$

$$a_{13} = s_x s_z (1 - c\theta) + s_y s\theta,$$

$$a_{21} = s_y s_x (1 - c\theta) + s_z s\theta,$$

$$\begin{aligned}
a_{22} &= (s_y^2 - 1)(1 - c\theta) + 1, \\
a_{23} &= s_y s_z (1 - c\theta) - s_x s\theta, \\
a_{31} &= s_z s_x (1 - c\theta) - s_y s\theta, \\
a_{32} &= s_z s_y (1 - c\theta) + s_x s\theta, \\
a_{33} &= (s_z^2 - 1)(1 - c\theta) + 1.
\end{aligned} \tag{1.37}$$

Equation (1.37) is called the *screw axis representation* of the orientation of a rigid body. This representation uses four parameters: three associated with the direction of the screw axis and one associated with the angle of rotation. However, only two of the three parameters associated with the direction of the screw axis are independent since they must satisfy the condition of a unit vector, $\mathbf{s}^T \mathbf{s} = 1$.

Given the screw axis and angle of rotation, we can compute the elements of the rotation matrix from Eq. (1.37). On the other hand, given a rotation matrix, we can compute the screw axis and the angle of rotation. The angle of rotation is obtained by summing the diagonal elements of the rotation matrix given by Eq. (1.37):

$$\theta = \cos^{-1} \frac{a_{11} + a_{22} + a_{33} - 1}{2}. \tag{1.38}$$

The direction of the screw axis is obtained by taking the differences between each pair of two opposing off-diagonal elements:

$$\begin{aligned}
s_x &= \frac{a_{32} - a_{23}}{2s\theta}, \\
s_y &= \frac{a_{13} - a_{31}}{2s\theta}, \\
s_z &= \frac{a_{21} - a_{12}}{2s\theta}.
\end{aligned} \tag{1.39}$$

From Eqs. (1.38) and (1.39) it appears that there are two solutions of the screw axis, one being the negative of the other. In reality, these two solutions represent the same screw, since a $-\theta$ rotation about the $-\mathbf{s}$ axis produces the same result as a $+\theta$ rotation about the $\mathbf{s}$ axis.

(c) Euler Angle Representations. The direction cosine representation of an orientation contains nine parameters, and the screw axis representation requires four. Since rotation is a motion with 3 degrees of freedom, a set of three independent parameters are sufficient to describe the orientation of a rigid body in space. Several sets of three-parameter representations have been reported in the literature. Perhaps, the most commonly used sets are the Euler

angles. In an Euler angle representation, three successive rotations about the coordinate axes of either a fixed coordinate system or a moving coordinate system are used to describe the orientation of a rigid body.

For convenience, we introduce three basic rotation matrices from which an Euler angle representation can be derived in terms of their product. When a rigid body performs a rotation of θ about the z -axis, $s_x = s_y = 0$ and $s_z = 1$. Hence the rotation matrix, Eq. (1.37), reduces to

$$R(z, \theta) = \begin{bmatrix} c\theta & -s\theta & 0 \\ s\theta & c\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}. \quad (1.40)$$

Similarly, when a rigid body performs a rotation of ψ about the x -axis, $s_y = s_z = 0$ and $s_x = 1$. Hence the rotation matrix, Eq. (1.37), reduces to

$$R(x, \psi) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & c\psi & -s\psi \\ 0 & s\psi & c\psi \end{bmatrix}, \quad (1.41)$$

and when a rigid body performs a rotation of ϕ about the y -axis, $s_x = s_z = 0$ and $s_y = 1$. Hence the rotation matrix, Eq. (1.37), reduces to

$$R(y, \phi) = \begin{bmatrix} c\phi & 0 & s\phi \\ 0 & 1 & 0 \\ -s\phi & 0 & c\phi \end{bmatrix}. \quad (1.42)$$

Roll–Pitch–Yaw Angles. We first consider three successive rotations of the moving frame B about the coordinate axes of the fixed frame A . Starting with the moving frame B coinciding with the fixed frame A , we rotate B about the x -axis by an angle ψ , resulting in an (u', v', w') system; followed by a second rotation of θ about the y -axis, resulting in an (u'', v'', w'') system; and then a third rotation of ϕ about the z -axis, resulting in the final (u, v, w) system, as shown in Fig. 1.26.

Since all rotations take place about the coordinate axes of a fixed frame, the resulting rotation matrix is obtained by premultiplying three basic rotation matrices:

$$\begin{aligned} R(\psi, \theta, \phi) &= R(z, \phi) R(y, \theta) R(x, \psi) \\ &= \begin{bmatrix} c\phi c\theta & c\phi s\theta s\psi - s\phi c\psi & c\phi s\theta c\psi + s\phi s\psi \\ s\phi c\theta & s\phi s\theta s\psi + c\phi c\psi & s\phi s\theta c\psi - c\phi s\psi \\ -s\theta & c\theta s\psi & c\theta c\psi \end{bmatrix}. \end{aligned} \quad (1.43)$$

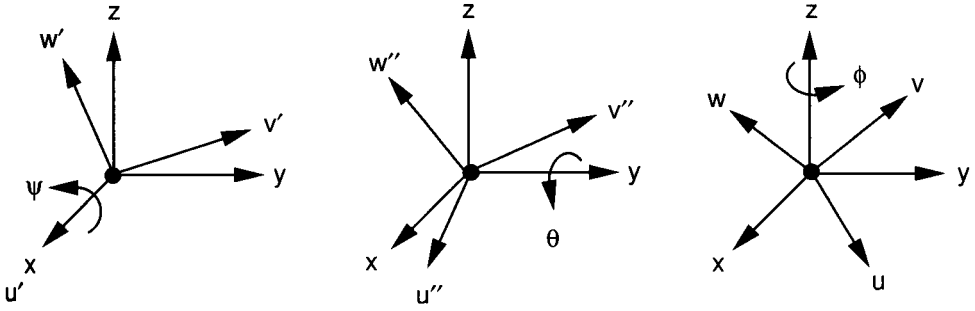


FIGURE 1.26. Successive rotations about the fixed coordinate axes.

The rotation about the x -axis is called a *roll*, the rotation about the y -axis is called a *pitch*, and the rotation about the z -axis is called a *yaw*. We call the convention of describing the orientation of a rigid body the *roll–pitch–yaw* angles representation. We note that successive rotations about the fixed coordinate axes result in a *premultiplication* of the matrices. Since finite rotations do not commute (Goldstein, 1980), the order of rotations cannot be exchanged arbitrarily.

Given the roll, pitch, and yaw angles, we can compute the overall rotation matrix from Eq. (1.43). On the other hand, given a rotation matrix, we can compute the equivalent roll–pitch–yaw angles as follows:

$$\begin{aligned}\theta &= \sin^{-1}(-a_{31}), \\ \psi &= \text{Atan2}(a_{32}/c\theta, a_{33}/c\theta), \\ \phi &= \text{Atan2}(a_{21}/c\theta, a_{11}/c\theta),\end{aligned}\tag{1.44}$$

provided that $c\theta \neq 0$, where $\sin^{-1} x$ is the arcsine function and $\text{Atan2}(y, x)$ is a two-argument arctangent function that yields one unique solution for the angle. Hence corresponding to a given rotation matrix, there are generally two possible solutions of the roll–pitch–yaw angles. However, if $\theta = \pm 90^\circ$, the solutions of Eq. (1.44) degenerate. For this special case, only the sum or the difference of ϕ and ψ can be computed.

w – u – w Euler Angles. Next, we consider three successive rotations of the rigid body about the coordinate axes of a moving frame B . Starting with the moving frame B coinciding with the fixed frame A , we rotate B about the body-attached w -axis by an angle ϕ , followed by a second rotation of θ about the rotated u' -axis, and then a third rotation of ψ about the rotated w'' -axis as shown in Fig. 1.27. We note that each rotation occurs about an axis whose location depends on the preceding rotations. The first rotation causes

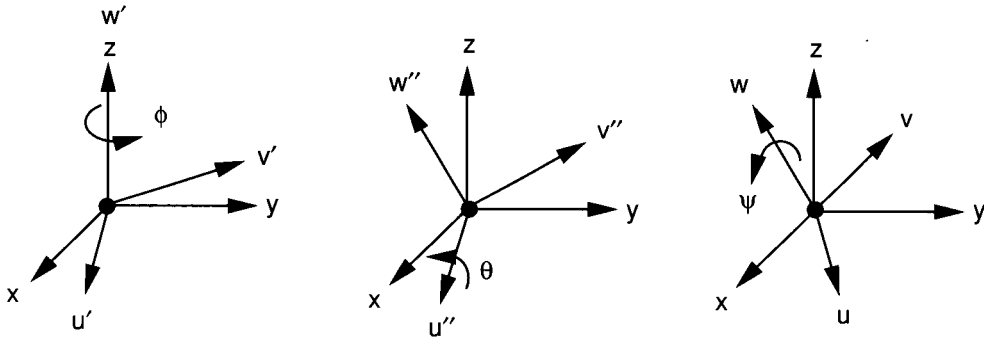


FIGURE 1.27. Successive rotations about the moving coordinate axes.

the (u, v, w) frame to move into the (u', v', w') location. The second rotation causes the (u', v', w') frame to move into the (u'', v'', w'') location. The third rotation causes the (u'', v'', w'') frame to rotate into the final (u, v, w) location. Three such successive rotations are called w – u – w or z – x – z Euler angles.

The resulting rotation matrix can be derived by a kinematic inversion, that is, by considering the orientation of frame A with respect to frame B . The inverse kinematics problem can be stated as a rotation of frame A about the w -axis by angle $-\phi$, followed by a second rotation of $-\theta$ about the u -axis, and followed by a third rotation of $-\psi$ about the w -axis. Under the kinematic inversion, the coordinate axes of frame B are considered as fixed. Hence the overall rotation matrix, ${}^B R_A(-\phi, -\theta, -\psi)$, can be written as

$${}^B R_A(-\phi, -\theta, -\psi) = {}^B R_A(w, -\psi) {}^B R_A(u, -\theta) {}^B R_A(w, -\phi). \quad (1.45)$$

Since ${}^A R_B = [{}^B R_A]^{-1}$ and ${}^B R_A^{-1}(w, -\phi) = {}^B R_A(w, \phi)$, we can expand Eq. (1.45) as follows:

$$\begin{aligned} {}^A R_B(\phi, \theta, \psi) &= [{}^B R_A(-\phi, -\theta, -\psi)]^{-1} \\ &= R(w, \phi) R(u, \theta) R(w, \psi) \\ &= \begin{bmatrix} c\phi c\psi - s\phi c\theta s\psi & -c\phi s\psi - s\phi c\theta c\psi & s\phi s\theta \\ s\phi c\psi + c\phi c\theta s\psi & -s\phi s\psi + c\phi c\theta c\psi & -c\phi s\theta \\ s\theta s\psi & s\theta c\psi & c\theta \end{bmatrix}. \end{aligned} \quad (1.46)$$

We note that successive rotations about the rotated coordinate axes of a moving frame result in a *postmultiplication* by the matrices. Given the w – u – w Euler angles, we can compute the resulting rotation matrix from

Eq. (1.46). On the other hand, given a rotation matrix, we can compute the w - u - w Euler angles as follows:

$$\begin{aligned}\theta &= \cos^{-1} a_{33}, \\ \phi &= \text{Atan2}(a_{13}/s\theta, -a_{23}/s\theta), \\ \psi &= \text{Atan2}(a_{31}/s\theta, a_{32}/s\theta),\end{aligned}\tag{1.47}$$

provided that $s\theta \neq 0$. When $\theta = 0$ or 180° , the solutions of Eq. (1.47) degenerates. For this degenerated case, only the sum or the difference of ϕ and ψ can be computed.

w - v - w Euler Angles. Another type of Euler angle representation consists of a rotation of angle ϕ about the w -axis, followed by a second rotation of θ about the rotated v -axis, and followed by a third rotation of ψ about the rotated w -axis. The resulting rotation matrix is obtained by a postmultiplication of three basic rotation matrices as follows:

$$\begin{aligned}{}^A R_B(\phi, \theta, \psi) &= R(w, \phi)R(v, \theta)R(w, \psi) \\ &= \begin{bmatrix} c\phi c\theta c\psi - s\phi s\psi & -c\phi c\theta s\psi - s\phi c\psi & c\phi s\theta \\ s\phi c\theta c\psi + c\phi s\psi & -s\phi c\theta s\psi + c\phi c\psi & s\phi s\theta \\ -s\theta c\psi & s\theta s\psi & c\theta \end{bmatrix}.\end{aligned}\tag{1.48}$$

Again, given the w - v - w Euler angles, we can compute the resulting rotation matrix from Eq. (1.48). On the other hand, given a rotation matrix, we can compute the w - v - w Euler angles as follows:

$$\begin{aligned}\theta &= \cos^{-1} a_{33}, \\ \phi &= \text{Atan2}(a_{23}/s\theta, a_{13}/s\theta), \\ \psi &= \text{Atan2}(a_{32}/s\theta, -a_{31}/s\theta),\end{aligned}\tag{1.49}$$

provided that $s\theta \neq 0$. When $\theta = 0$ or 180° , the solutions of Eq. (1.49) degenerates. In this case, only the sum or the difference of ϕ and ψ can be computed.

1.5.3 Description of a Location

As pointed out earlier, the location of the rigid body can be described by the position of the origin Q and the orientation of the moving frame with respect to the fixed frame. Figure 1.22 shows that the position of a point P of the rigid body can be expressed in the fixed frame A as ${}^A \mathbf{p} = \overline{OP}$. It can also be

expressed in the moving frame B as ${}^B\mathbf{p} = \overline{QP}$. To derive a relation between ${}^A\mathbf{p}$ and ${}^B\mathbf{p}$, we construct the vector $\overline{OP}$ as a sum of two vectors:

$$\overline{OP} = \overline{OQ} + \overline{QP}, \quad (1.50)$$

where $\overline{OQ} = {}^A\mathbf{q}$ denotes the position of Q with respect to the fixed frame A .

Let the orientation of the moving frame B with respect to the fixed frame A be defined by the rotation matrix AR_B . Then Eq. (1.50) can be written as

$${}^A\mathbf{p} = {}^AR_B {}^B\mathbf{p} + {}^A\mathbf{q}. \quad (1.51)$$

Equation (1.51) describes the position of a point in a rigid body in terms of the position of the origin Q and the orientation of the moving frame B with respect to the fixed frame A . The leading superscript of ${}^B\mathbf{p}$ cancels with the trailing subscript of AR_B , leaving all quantities as vectors expressed in the fixed frame A .

Since we assume that initially, the moving frame coincides with the fixed frame, we may consider ${}^B\mathbf{p}$ as the first position of a point P and ${}^A\mathbf{p}$ as the second position of the same point expressed in the fixed frame A . The first term on the right-hand side of Eq. (1.51) represents the contribution due to a rotation of the rigid body about some axis, and the second term represents the contribution due to a translation along the direction of ${}^A\mathbf{q}$. We observe that the general spatial displacement of a rigid body can be considered as a rotation plus a translation. This is well known as *Chasles' theorem*.

1.6 HOMOGENEOUS TRANSFORMATIONS

Equation (1.51) provides a general transformation of a position vector from the moving frame to the fixed frame. The first term on the right-hand side of the equation represents the contribution due to a rotation and the second term that due to a translation of the moving frame respect to the fixed frame. The equation is not in a compact form, because the 3×3 rotation matrix does not provide for the translation. To write Eq. (1.51) in a better-appearing form, we introduce the concepts of homogeneous coordinates and homogeneous transformation matrix.

1.6.1 Homogeneous Coordinates

Let $\mathbf{p} = [p_x, p_y, p_z]^T$ be the position vector of a point with respect to a reference frame A in three-dimensional space. We define the *homogeneous*

coordinates of $\mathbf{p}$ as

$$\hat{\mathbf{p}} \equiv [\rho p_x, \rho p_y, \rho p_z, \rho]^T. \quad (1.52)$$

Thus the homogeneous coordinates of a point $\mathbf{p}$ in frame A are represented by a vector $\hat{\mathbf{p}}$ in a four-dimensional space. The fourth coordinate ρ is a nonzero *scaling factor*. In general, an N -dimensional position vector becomes an $(N+1)$ -dimensional vector in a homogeneous coordinate system. The concept of homogeneous coordinates is useful in developing matrix transformations that include rotation, translation, scaling, and perspective transformation (Ballard and Brown, 1982).

From the definition above, we see that a three-dimensional vector can be recovered from its four-dimensional homogeneous coordinates by dividing the first three homogeneous coordinates by the fourth coordinate; that is,

$$p_x = \frac{\hat{p}_x}{\rho}, \quad p_y = \frac{\hat{p}_y}{\rho}, \quad \text{and} \quad p_z = \frac{\hat{p}_z}{\rho}. \quad (1.53)$$

We note that the homogeneous coordinates $\hat{\mathbf{p}}$ are not unique, since any nonzero scaling factor ρ will yield the same three-dimensional vector $\mathbf{p}$. For example, $\hat{\mathbf{p}}_1 = [\rho_1 p_x, \rho_1 p_y, \rho_1 p_z, \rho_1]^T$ and $\hat{\mathbf{p}}_2 = [\rho_2 p_x, \rho_2 p_y, \rho_2 p_z, \rho_2]^T$ represent the same position vector $\mathbf{p} = [p_x, p_y, p_z]^T$ in a three-dimensional space. For the kinematics of mechanisms and robot manipulators, we choose a scaling factor of $\rho = 1$ for convenience. When the scaling factor is set to unity, the first three homogeneous coordinates represent the actual coordinates of a three-dimensional vector. Hence the position vector of a point is given simply as

$$\hat{\mathbf{p}} = [p_x, p_y, p_z, 1]^T. \quad (1.54)$$

In subsequent chapters, we omit the *hats* from the equation for brevity.

1.6.2 Homogeneous Transformation Matrix

The *homogeneous transformation matrix* is a 4×4 matrix that is defined for the purpose of mapping a homogeneous position vector from one coordinate system into another. The matrix can be partitioned into four submatrices as follows:

$${}^A T_B = \begin{bmatrix} {}^A R_B(3 \times 3) & \vdots & {}^A \mathbf{q}(3 \times 1) \\ \dots\dots\dots & \vdots & \dots\dots\dots \\ \boldsymbol{\gamma}(1 \times 3) & \vdots & \rho(1 \times 1) \end{bmatrix}. \quad (1.55)$$

The upper left 3×3 submatrix ${}^A R_B$ denotes the *orientation* of a moving frame B with respect to a reference frame A , the upper right 3×1 submatrix ${}^A \mathbf{q}$ denotes the *position* of the origin of the moving frame relative to the fixed frame, the lower left 1×3 submatrix γ represents a *perspective transformation*, and the lower right element ρ is a *scaling factor*. For kinematics of mechanisms and robot manipulators, the scaling factor is set to unity and the 1×3 perspective transformation matrix is set to zero. See Nevatia (1982) for a more detailed description of the matrix.

Using the homogeneous coordinates, Eq. (1.51) can be written in the following compact form:

$${}^A \hat{\mathbf{p}} = {}^A T_B {}^B \hat{\mathbf{p}}, \quad (1.56)$$

where

$${}^A T_B = \begin{bmatrix} {}^A R_B & \vdots & {}^A \mathbf{q} \\ \dots\dots & \vdots & \dots\dots \\ 0 & 0 & 0 & \vdots & 1 \end{bmatrix},$$

$${}^A \hat{\mathbf{p}} = [p_x, p_y, p_z, 1]^T,$$

$${}^B \hat{\mathbf{p}} = [p_u, p_v, p_w, 1]^T.$$

For example, the homogeneous transformation matrix for a simple rotation about the z -axis is given by

$${}^A T_B(z, \theta) = \begin{bmatrix} c\theta & -s\theta & 0 & 0 \\ s\theta & c\theta & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}. \quad (1.57)$$

The transformation matrix for a pure translation is given by

$${}^A T_B(\mathbf{q}) = \begin{bmatrix} 1 & 0 & 0 & q_x \\ 0 & 1 & 0 & q_y \\ 0 & 0 & 1 & q_z \\ 0 & 0 & 0 & 1 \end{bmatrix}. \quad (1.58)$$

In summary, Eq. (1.56) provides a transformation of homogeneous coordinates from one frame to another. Given the transformation matrix ${}^A T_B$, we can locate the moving frame B with respect to the fixed frame A ; and given the location of a moving frame B with respect to a fixed frame A , we can find the transformation matrix ${}^A T_B$.

Although the transformation matrix ${}^A T_B$ is not orthogonal (i.e., ${}^A T_B^{-1} \neq {}^A T_B^T$), its inverse transformation does exist. Multiplying both sides of

Eq. (1.51) by ${}^A R_B^{-1}$ and making use of the fact that ${}^A R_B^{-1} = {}^A R_B^T$, we obtain

$${}^B \mathbf{p} = {}^A R_B^T {}^A \mathbf{p} - {}^A R_B^T {}^A \mathbf{q}. \quad (1.59)$$

Using the homogeneous coordinates, Eq. (1.59) can be written as

$${}^B \hat{\mathbf{p}} = {}^A T_B^{-1} {}^A \hat{\mathbf{p}}. \quad (1.60)$$

Hence

$${}^A T_B^{-1} = {}^B T_A = \begin{bmatrix} {}^A R_B^T & \vdots & -{}^A R_B^T {}^A \mathbf{q} \\ \dots\dots & \vdots & \dots\dots\dots \\ 0 & 0 & 0 & \vdots & 1 \end{bmatrix}. \quad (1.61)$$

1.6.3 Composite Homogeneous Transformation

Homogeneous transformation matrices can be multiplied together to obtain a composite transformation matrix. However, special attention should be paid to the order of multiplication since finite rotations are not commutative. The problem is further complicated, since a rigid body can rotate about the coordinate axes of either a fixed reference frame or the body-attached moving frame. The following rules are helpful for the determination of a composite transformation matrix:

1. At the initial position, the moving frame B and the fixed frame A are coincident. Hence $T = I$ is an identity matrix.
2. Rotation and translation about the coordinate axes of a fixed frame results in a premultiplication of the two matrices.
3. Rotation and translation about the coordinate axes of a moving frame results in a postmultiplication of the two matrices.

1.7 MECHANICS OF ROBOT MANIPULATORS

Robotics is the science or study of the basic technologies associated with the theory and application of robotic systems. The study involves both theoretical and applied research, which can be divided into manipulator design, basic mechanics, trajectory planning and control, programming and machine intelligence, and so on. *Mechanics* is a branch of science that deals with energy and forces and their effect on the motion of a mechanical system. The study involves three interrelated subjects: kinematics, statics, and dynamics.

1.7.1 Kinematics

Kinematics deals with the aspects of motion without regard to the forces and/or torques that cause it. The science of kinematics deals with the position, velocity, acceleration, and higher-order derivatives of the position variables with respect to time or other variables. Hence kinematics is concerned only with the geometrical and time properties of a motion. The joint variables of a robot manipulator are related to the position and orientation of the end effector by the constraints imposed by the joints. These kinematic relations are the focal points of interest in a study of the kinematics of robot manipulators. The study can be approached from two different points of view: kinematic analysis and kinematic synthesis. However, the processes of kinematic analysis and kinematic synthesis are intertwined. A designer needs a skillful analysis ability to evaluate various design alternatives under consideration and to arrive at the best design. Hence a better understanding of the kinematics is the first concern in the design and control of robot manipulators.

Kinematic analysis deals with the derivation of relative motions among various links of a given manipulator. There are two types of kinematic analysis problems: direct kinematics and inverse kinematics. In the programming of a robot manipulator, typically a set of desired positions and orientations, and perhaps the time derivatives of the positions and orientations of the end effector, are specified in space. The problem is to find all possible sets of actuated joint variables and their corresponding time derivatives which will bring the end effector to the set of desired positions and orientations with the desired motion characteristics. This is known as *inverse kinematics*. On the other hand, sometimes the actuated joint variables and possibly their time derivatives are obtained from readings of sensors installed at the joints, from which we wish to find all possible sets of end effector positions and orientations and their corresponding time derivatives. This is called *direct kinematics* or *forward kinematics*. Both direct and inverse kinematics problems can be solved by various methods of analysis, such as geometric vector analysis, matrix algebra, screw algebra, and so on (Erdman and Sandor, 1991; Kimbrell, 1991; Lange, 1995; Martin, 1982).

Kinematic synthesis is the reverse process of kinematic analysis. In this case, a designer is challenged to devise a new manipulator or machine that will possess certain desired kinematic properties. Specifically, given a set of desirable positions and orientations of the end effector and possibly their time derivatives in space, the corresponding actuated joint variables and the type and geometry of a manipulator are to be determined. Some researchers have focused on the synthesis of manipulators with the workspace as an optimality criterion. The kinematic synthesis problem can be further divided into three interrelated phases: *type synthesis*, *number synthesis*, and *dimensional syn-*

thesis. See Erdman and Sandor (1991) or Hartenberg and Denavit (1964) for a description of these three phases of kinematic synthesis.

1.7.2 Statics

Statics deals with the relations of forces that produce equilibrium among the various members of a robot manipulator. A manipulator may be acted upon by forces that arise from various sources, such as forces of gravity, forces of applied loads, frictional forces, inertia forces, and so on. These forces must be considered carefully during the design stage of a robot manipulator so that its parts can be sized properly and the manipulator will function correctly. Obviously, inertia forces are excluded from the static force analysis. The forces of equilibrium depend on the *configuration* or *posture* of a robot manipulator and are not time dependent.

1.7.3 Dynamics

Dynamics deals with the forces and/or torques required to cause the motion of a system of bodies. The study includes inertia forces as one of the principal concerns. The dynamics of a robot manipulator is a very complicated subject. Typically, the end effector is to be guided through a given path with certain prescribed motion characteristics. A set of torque and/or force functions must be applied at the actuated joints in order to produce that motion. These actuating torque and/or force functions depend not only on the spatial and temporal attributes of the given path but also on the mass properties of the links, the payload, the externally applied forces, and so on.

The dynamics of robot manipulators can also be approached from two different points of view: dynamical analysis and dynamical synthesis.

Dynamical analysis deals with derivation of the equations of motion of a given manipulator. There are two types of dynamical analysis problems: direct dynamics and inverse dynamics. *Direct dynamics* can be defined as follows: Given a set of actuated joint torque and/or force functions, calculate the resulting motion of the end effector as a function of time; and *inverse dynamics* as: Given a trajectory of the end effector as a function of time, find a set actuated joint torque and/or force functions which will produce that motion. The computational efficiency of direct dynamics is not as critical since it is used primarily for computer simulations of a robot manipulator. On the other hand, an efficient inverse dynamical model becomes extremely important for real-time, model-based control of a robot manipulator. Various methods of analysis, such as the Newton–Euler equations, the Lagrangian equations of motion, and the principle of virtual work can be applied for the dynamical analysis of robot manipulators.

Dynamical synthesis is the reverse of dynamical analysis. Specifically, a new manipulator is to be designed with certain desired dynamical motion characteristics. The dynamical synthesis problem is much more complicated than the dynamical analysis problem and is not treated in this book, which concentrates on recent advances in the kinematic, static, and dynamical analysis of robot manipulators. Both serial and parallel manipulators are covered. A thorough understanding of the mechanics of robot manipulators is a prerequisite for achieving better design, path planning, and control of such electromechanical systems. Readers of this book are strongly encouraged to consult other textbooks on robot sensing, control, path planning, programming languages, and machine intelligence (e.g., Asada and Slotine, 1986; Craig, 1986; Fu et al., 1987; Paul, 1981; Rivin, 1987; Schilling, 1990).

1.8 SUMMARY

We first gave a brief description of the background and historical development of robot manipulators. Then we provided a review of the definitions of links, joints, kinematic chains, mechanisms, and machines. The concepts of degrees of freedom and the loop-mobility criterion of a mechanism were introduced. The essential components of a robotic system were described. Robot manipulators were classified according to their degrees of freedom, kinematic structure, drive technology, workspace envelope, and motion characteristics. Several industrial robots were used to demonstrate the state-of-the-art development. Furthermore, descriptions of the position, orientation, and location of a rigid body with respect to a fixed frame were presented. Several different methods of specifying the orientation of a rigid body with respect to a fixed frame were studied. Finally, the concepts of homogeneous coordinates, homogeneous transformation, and composite homogeneous transformation were introduced. These fundamental concepts will be useful for the development of the mechanics of robot manipulators in subsequent chapters.

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EXERCISES

1. Find the number of degrees of freedom associated with the planar manipulator shown in Fig. 1.28, where a moving platform is shown connected to a fixed base by three limbs. Each limb consists of two binary links and three parallel revolute joints. Is this a serial or a parallel manipulator?

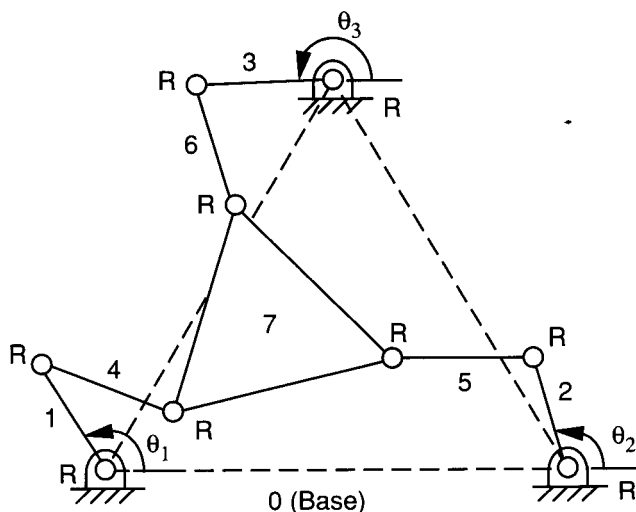


FIGURE 1.28. Planar 3RRR manipulator.

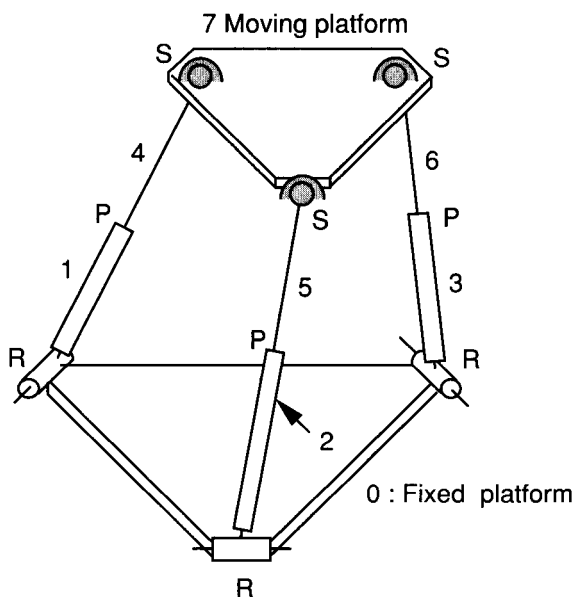


FIGURE 1.29. Spatial 3RPS manipulator.

2. Figure 1.29 shows a spatial mechanism in which a moving platform is connected to a fixed base by three extensible limbs with a spherical joint at the upper end and a revolute joint at the lower end. Furthermore, each limb is made up of two binary links connected by a prismatic joint. Find the number of degrees of freedom of the mechanism.
3. Figure 1.30 shows a moving platform that is connected to a fixed base by three limbs with a universal joint on each end. Each limb is made up of two binary links connected by a prismatic joint. Find the degrees of freedom associated with this spatial manipulator.
4. A rigid body is first rotated about the fixed x -axis by 45° and then about the fixed y -axis by 30° . Find the resulting rotation matrix.
5. A rigid body is first rotated about the body-attached w -axis by 30° and then about the rotated v -axis by 45° . Find the resulting rotation matrix.
6. What is the resulting rotation matrix for a rotation of 60° about the fixed x -axis, followed by a rotation of 45° about the y -axis, followed by another rotation of 30° about the z -axis?

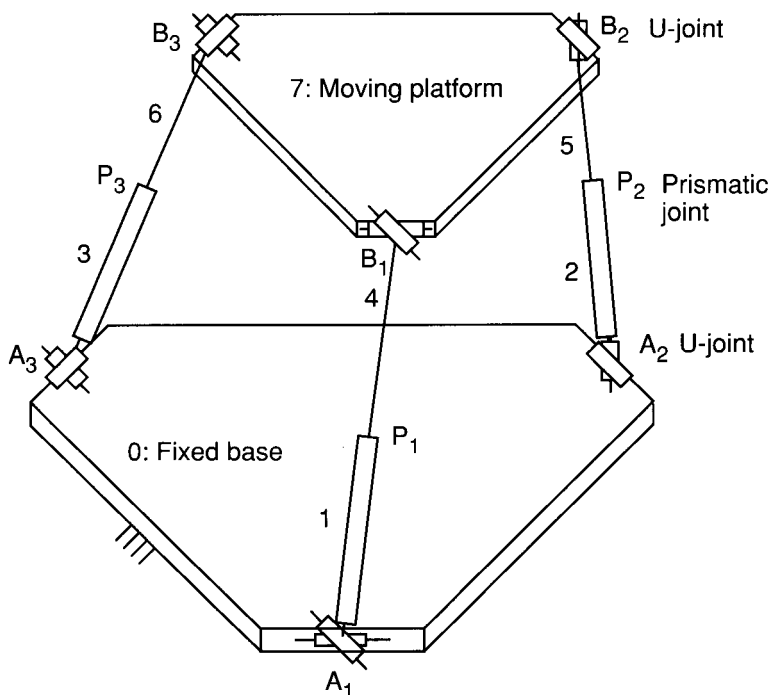


FIGURE 1.30. Spatial 3UPU manipulator.

7. Find the axis of rotation associated with the rotation matrix

$${}^A R_B = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix},$$

8. Find the equivalent screw axis and the angle of rotation associated with the rotation matrix

$${}^A R_B = \begin{bmatrix} 0.866 & -0.500 & 0.000 \\ 0.500 & 0.866 & 0.000 \\ 0.000 & 0.000 & 1.000 \end{bmatrix}.$$

9. Find the equivalent x - y - z roll-pitch-yaw angles associated with the rotation matrix

$${}^A R_B = \begin{bmatrix} 1.000 & 0.000 & 0.000 \\ 0.000 & 0.866 & -0.500 \\ 0.000 & 0.500 & 0.866 \end{bmatrix}.$$

10. Find the equivalent w - u - w Euler angles associated with the rotation matrix

$${}^A R_B = \begin{bmatrix} 0.866 & -0.500 & 0.000 \\ 0.433 & 0.750 & -0.500 \\ 0.250 & 0.433 & 0.866 \end{bmatrix}.$$

11. A rigid body is rotated about the fixed z -axis by 30° followed by a translation of 2 units in the x direction and 4 units in the y -direction. Find the transformation matrix ${}^A T_B$ and its inverse transformation, ${}^B T_A$.